Q1

An ideal gas with pressure P, volume V and temperature T is expanded isothermally to a volume 2V and final pressure P_I. The same gas is expanded adiabatically to a volume 2V and final pressure in this case is P_A. In terms of the ratio of the two specific heats for the gas γ, the ratio P_I/P_A is

(1) $2^{\gamma-1}$

(2) $2^{1-\gamma}$

(3) 2^{γ}

(4) 2γ

Q2

When only carrier is sent, antenna current of an AM transmitter is 8A but the current increases to 8.88A when the carrier is sinusoidally modulated. The percentage of modulation is

(1) 62%

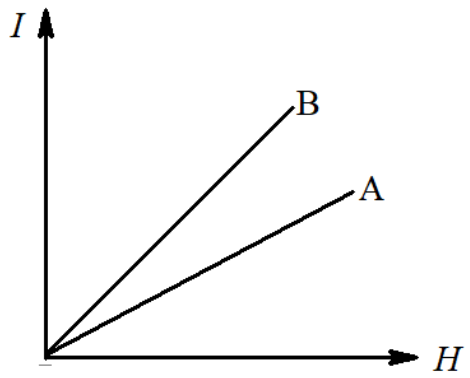
(2) 65%

(3) 68%

(4) 71%

Q3

The following figure shows the variation of intensity of magnetisation I versus the applied magnetic field intensity H, for two magnetic materials A and B



Which of the material have a larger susceptibility for a given field at constant temperature?

(1) Material B

(2) Material B and Material A have the same susceptibility

(3) Material A

(4) None of these

Q4

The pressure of a medium is changed from 1.01×10^5 Pa to 1.165×10^5 Pa and change in volume is 10% keeping temperature constant.The bulk modulus of the medium is

(1) 204.8×10^5 Pa

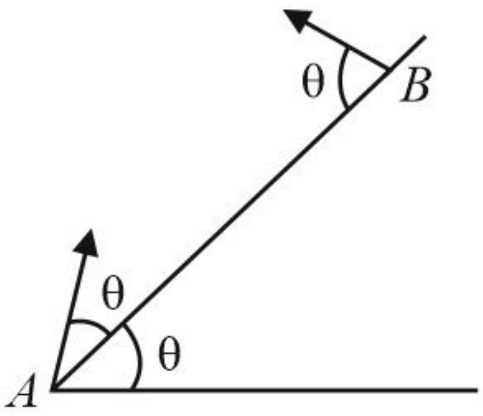
(2) 102.4×10^5 Pa

(3) 51.2×10^5 Pa

(4) 1.55×10^5Pa

Q5

From an inclined plane, two particles are projected with the same speed at same angle θ, one up and other down the plane as shown in the figure. Which of the following statement (s) is/are correct?



(1) The time of flight of each particle is the same.

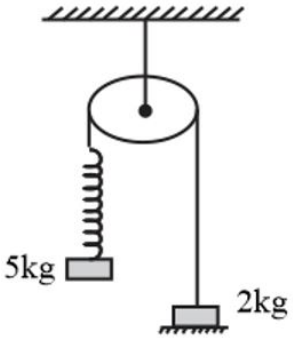
(2) The particles will collide the plane with same speed

(3) Both the particles strike the plane perpendicularly

(4) The particles will collide in mid air if projected simultaneously and time of flight of each particle is less than the time of collision

Q6

The system shown in the figure is released from rest with spring in natural length and string just taut. The pulley and the spring are massless and friction is absent everywhere. The value of spring constant is $k = 40\ N\ m^{-1}$. The speed of $5\ kg$ block when $2\ kg$ block leaves contact with the support below it, will be



(1) $\sqrt{2}\ m\ s^{-1}$

(2) $2\sqrt{2}\ m\ s^{-1}$

(3) $2\ m\ s^{-1}$

(4) $4\sqrt{2}\ m\ s^{-1}$

Q7

Two factories are sounding their sirens at $800\ Hz$. A man goes from one factory to other at a speed of $2\ m/s$. The velocity of sound is $320\ m/s$. The number of beats heard by the person in one second will be

(1) 10

(2) 4

(3) 2

(4) 8

Q8

A positive charge q is given to each plate of a parallel plate air capacitor having plate area A and plate separation d, then

(1) since both the plates are identically charged, therefore, capacitance becomes equal to zero

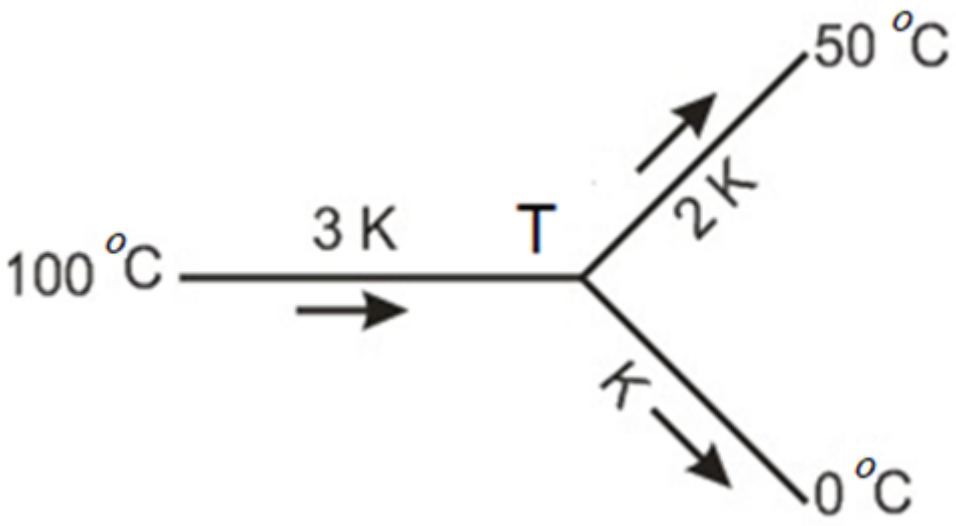
(2) energy stored in the space between the capacitor plates is equal to $\frac{q^2}{\varepsilon_0 A^2}$

(3) no charge appears on inner surface of the plates

(4) potential difference between the plates is equal to $\frac{2qd}{\varepsilon_0 A}$

Q9

Three rods of same dimension have thermal conductivity 3 K, 2 K and K as shown in figure.



Then, the temperature of the junction in steady state is

(1) $\frac{200}{3}\,^{o}\mathrm{C}$

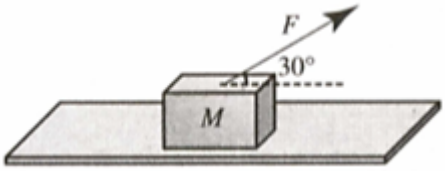
(2) $\frac{100}{3}\,^{o}\mathrm{C}$

(3) $75\,^{o}\mathrm{C}$

(4) $\frac{50}{3}\,^{o}\mathrm{C}$

Q10

A block of mass, $M = 5$ kg is resting on a rough horizontal surface for which the coefficient of friction is 0.2. Let a force, $F = 40$ N be applied, what will then the acceleration of the block be?



(1) $5.73\ \mathrm{m\ s^{-2}}$

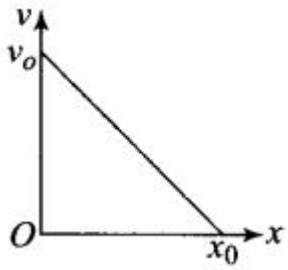
(2) $8.0\ \mathrm{m\ s^{-2}}$

(3) $3.17\ \mathrm{m\ s^{-2}}$

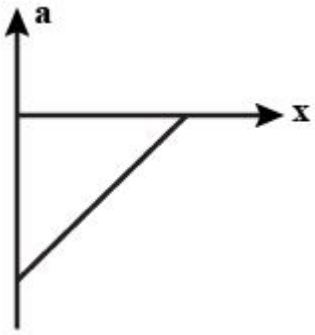
(4) $10.0\ \mathrm{m\ s^{-2}}$

Q11

The velocity displacement graph of a particle moving along a straight line is -

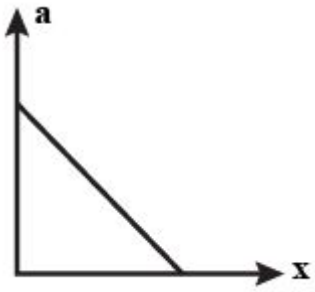


The most suitable acceleration-displacement graph will be

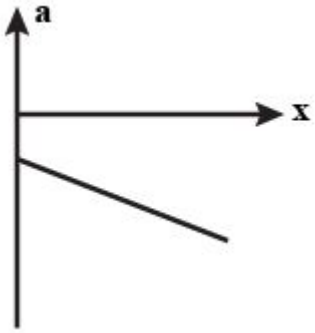


(1)

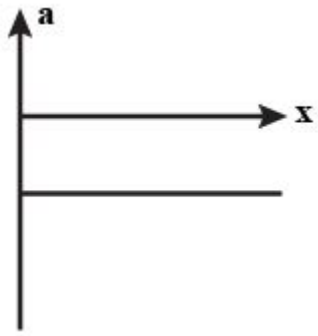
(2)



(3)



(4)



Q12

Consider the nuclear reaction $\mathrm{X}^{200} \rightarrow \mathrm{A}^{110} + \mathrm{B}^{80} + 10\mathrm{n}^{1}$. If the binding energy per nucleon for X, A and B are $7.4\ \text{MeV}$, $8.2\ \text{MeV}$ and $8.1\ \text{MeV}$ respectively, then the energy released in the reaction -

(1) $70\ \text{MeV}$

(2) $200\ \text{MeV}$

(3) $190\ \text{MeV}$

(4) $10\ \text{MeV}$

Q13

If force (F), length (L) and time (T) be considered fundamental units, then the units of mass will be

(1) $\left[\mathrm{F\ L\ T^{-2}}\right]$

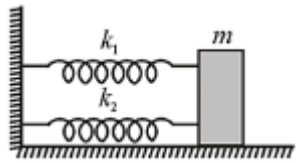
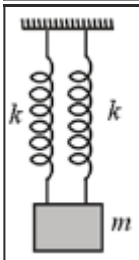
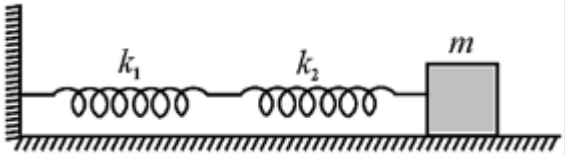
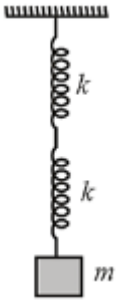
(2) $\left[\mathrm{F\ L^{-2}T^{-1}}\right]$

(3) $\left[\mathrm{F\ L^{-1}T^{2}}\right]$

(4) $\left[\mathrm{F^{2}L\ T^{-2}}\right]$

Q14

Match the combination of springs shown in Column I with their respective time periods in Column II.

	Column - I		Column - II
(a)		(i)	$T = 2\pi\sqrt{\frac{m(k_1+k_2)}{k_1k_2}}$
(b)		(ii)	$T = 2\pi\sqrt{\frac{2m}{k}}$
(c)		(iii)	$T = 2\pi\sqrt{\frac{m}{2k}}$
(d)		(iv)	$T = 2\pi\sqrt{\frac{m}{k_1+k_2}}$

(1) $a-(i),\ b-(ii),\ c-(iv),\ d-(iii)$

(2) $a-(iv),\ b-(iii),\ c-(i),\ d-(ii)$

(3) $a-(iii),\ b-(i),\ c-(iv),\ d-(ii)$

(4) $a-(i),\ b-(iii),\ c-(ii),\ d-(iv)$

Q15

In the following a statement of Assertion is followed by a statement of Reason.

Assertion: An armature coil is rotating with constant angular speed in uniform magnetic field. Magnitude of induced emf in coil is maximum when field lines are parallel.

Reason: Induced electric field is produced by non-uniform steady magnetic field.

(1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion.

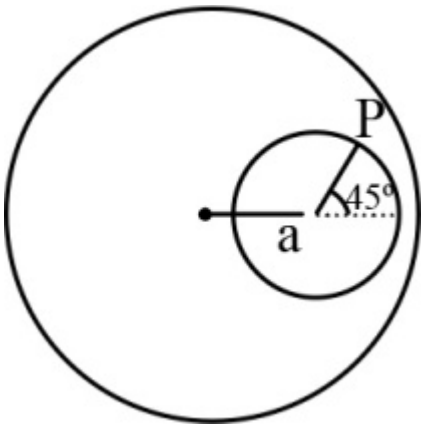
(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

Q16

From a sphere of uniform volume charge density ρ, a cavity of radius r is made as shown in the figure. An electron (charge e, mass m) is released from rest inside the cavity from point P. The centre of sphere and centre of cavity are separated by a distance a. The time after which the electron again touches the sphere is -



(1) $\sqrt{\frac{6\sqrt{2}r\varepsilon_0 m}{e\rho a}}$

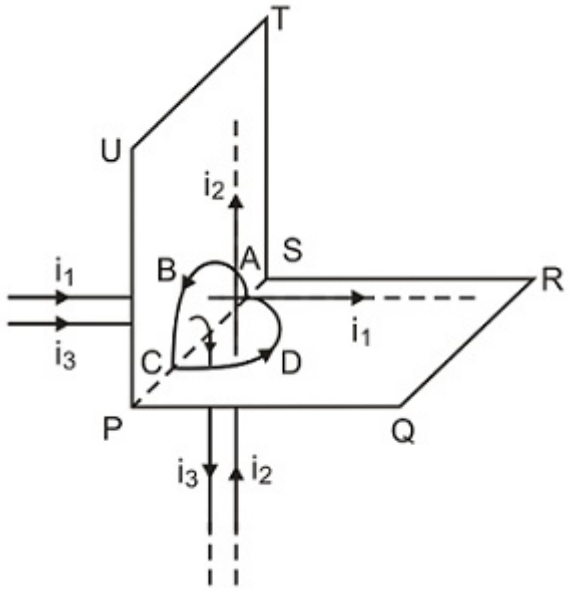
(2) $\sqrt{\frac{\sqrt{2}r\varepsilon_0 m}{6\rho a}}$

(3) $\sqrt{\frac{6r\varepsilon_0 m}{e\rho a}}$

(4) $\sqrt{\frac{r\varepsilon_0 m}{e\rho a}}$

Q17

Figure shows an amperian path ABCDA. Part ABC is in vertical plane PSTU while part CDA is in horizontal plane PQRS. Direction of circulation along the path is shown by an arrow near point B and at D. $\oint \vec{B} \cdot d\vec{l}$ for this path according to Ampere's law will be



(1) $(i_1 - i_2 + i_3)\mu_0$

(2) $(-i_1 + i_2)\mu_0$

(3) $i_3\mu_0$

(4) $(i_1 + i_2)\mu_0$

Q18

Assertion: In parallel combination of electrical appliances, total power consumption is equal to the sum of the powers of the individual appliances.

Reason: In parallel combination, the voltage across each appliance is the same, as required for the proper working of electrical appliance.

(1) Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but the Reason is false.

(4) Assertion is false but Reason is true.

Q19

Match List- I (Experiment performed) with List- II (Phenomena discovered/associated)and select the correct option from the options given below the lists

	List-I		List-II
(a)	Davisson and Germer experiment	(i)	Wave nature of electrons
(b)	Millikan's oil drop experiment	(ii)	Charge of an electron
(c)	Rutherford experiment	(iii)	Quantisation of energy levels
(d)	Franck-Hertz experiment	(iv)	Existence of the nucleus

(1) (a)- (iii), (b)- (iv), (c)- (i), (d)- (ii)

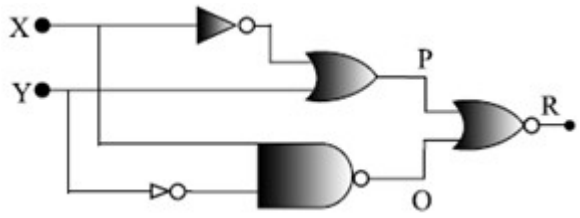
(2) (a)- (i), (b)- (ii), (c)- (iv), (d)- (iii)

(3) (a)- (iv), (b)- (iii), (c)- (ii), (d)- (i)

(4) (a)- (i), (b)- (ii), (c)- (iii), (d)- (iv)

Q20

The figure gives a system of logic gates. From the study of the truth table, it can be found that to produce a high output (1) at R, we must have



(1) $X = 0, Y = 1$

(2) $X = 1, Y = 1$

(3) $X = 1, Y = 0$

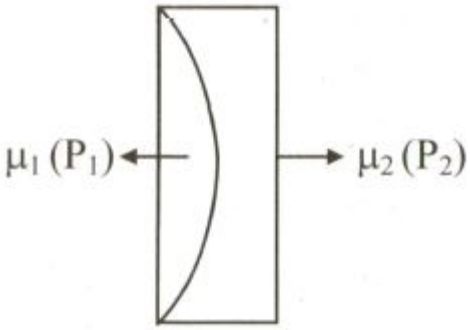
(4) $X = 0, Y = 0$

Q21

Molecules of hydrogen and helium are kept at temperatures $27\ ^{\circ}\mathrm{C}$ and $327\ ^{\circ}\mathrm{C}$ respectively. What will be their ratio of de Broglie wavelengths?

Q22

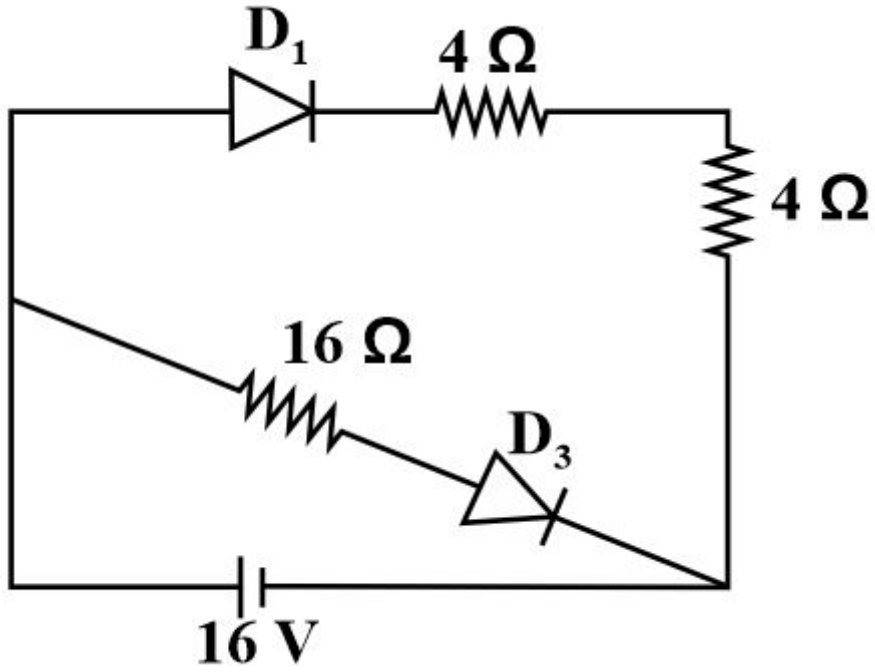
A plano-convex lens with $\mathrm{R.I.}\ \mu_1$ and power P_1 is placed in contact with a plano concave lens with $\mathrm{R.I.}\ \mu_2$ and power P_2.



Let both lenses have same radius of curvature R each and $\mathrm{P_2 = 2P_1}$. Then the refractive indices are related to each other as $\mu_1 + \frac{\mu_2}{2} = \mathrm{n}$. What is the value of $2\mathrm{n}$?

Q23

In the given circuit, what is the current (in A) through the battery?



Q24

Power supplied to a particle of mass 2 kg varies with time as $\mathrm{P} = \frac{3\mathrm{t}^2}{2}$ W, where t is in seconds. If the velocity of the particle at $\mathrm{t} = 0$ is $\mathrm{v} = 0$, then what is the velocity (in $\mathrm{m\ s^{-1}}$) of the particle at $\mathrm{t} = 2$ s?

Q25

An ac circuit has an inductor and a resistor of resistance R in series, such that $X_L = 3R$. Now, a capacitor is added in series such that $X_C = 2R$. the ratio of the new power factor with the old power factor of the circuit is $\sqrt{5} : x$. The value of x is

Q26

The wavelength of light used in a biprism experiment is $6000\ \overset{o}{A}$ and I_0 is the intensity of central maximum. If distance between two slits is 0.25 cm and the distance of the screen from slits is 120 cm then at what distance (in μm) from the central maxima, the intensity will be $\frac{\mathrm{I}_0}{2}$?

Q27

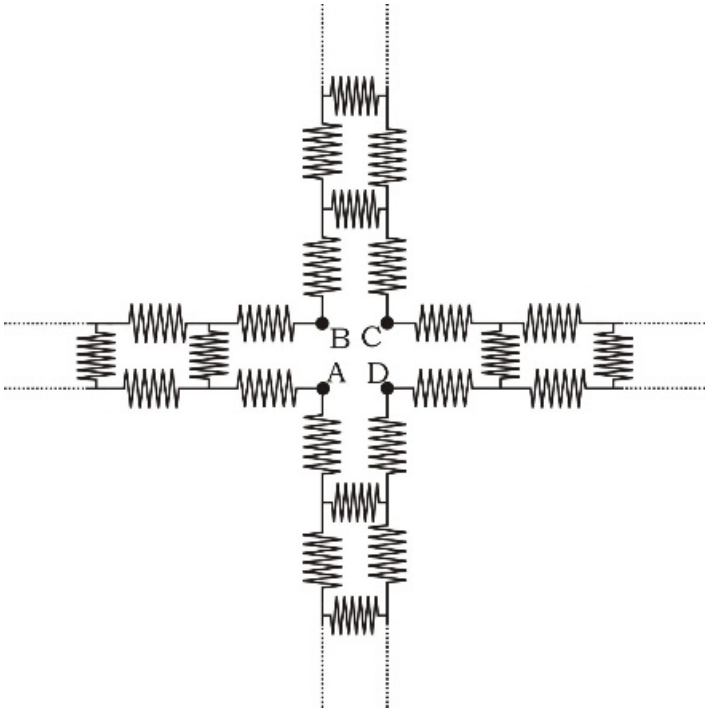
What is the percentage change in the de-Broglie wavelength of the particle if its kinetic energy is increased by 15 times?

Q28

A point source of light of power P_0 is placed at a distance of 4 m from the centre of a thin hemispherical shell as shown in the figure, The shell has a radius of 3 m and it behaves like a perfect black body. If the temperature of the hemisphere is related to the power of the source as $T^4 = \frac{P_0}{n\pi\sigma}$, where σ is the Stefan's constant, then find the value of $\frac{n}{10}$.

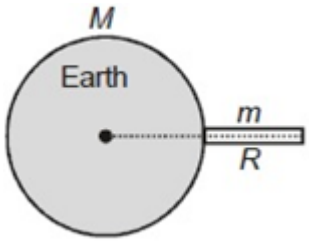
Q29

Four infinite ladder alike networks are combined as shown in figure .Each resistance is $R\,\Omega$. The equivalent resistance between A and B is R_{AB} and between A and C is R_{AC}. Let ratio $\frac{R_{\mathrm{AB}}}{R_{\mathrm{AC}}}$ is $\frac{x}{4}$ then find the value of x.



Q30

The gravitational force between earth of radius R and mass M and rod of length R and mass m placed as shown in the figure is $\frac{12GMm}{nR^2}$. Find the value of n.



Q31

Take two solutions of glucose. When 1 L of the first solution is mixed with V L of the second solution, the osmotic pressure of the resultant solution becomes 2.5 atm. Find the volume of the second solution if two solutions have osmotic pressures as 1.0 and 3.5 atm, respectively.

(1) 1.0 L

(2) 1.5 L

(3) 2.5 L

(4) 3.5 L

Q32

Consider the following statement :

(I) $CH_3\overset{\oplus}{O}CH_2$ is more stable than $CH_3\overset{\oplus}{C}H_2$

(II) $Me_3\overset{\oplus}{C}$ is more stable than $CH_3CH_2\overset{\oplus}{C}H_2$

(III) $CH_2=CH-\overset{\oplus}{C}H_2$ is more stable than $CH_3CH_2\overset{\oplus}{C}H_2$

(IV) $CH_2=\overset{\oplus}{C}H$ is more stable than $CH_3\overset{\oplus}{C}H_2$

Of these statement:

(1) I and II are correct

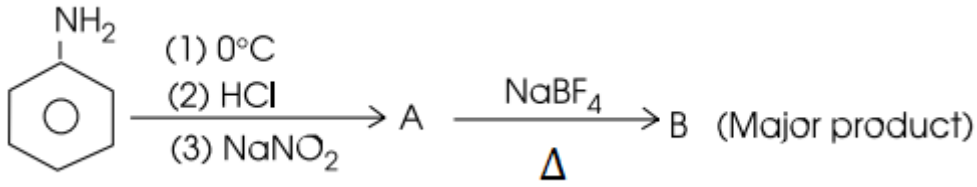
(2) III and IV are correct

(3) I, II and III are correct

(4) II, III and IV are correct

Q33

Determine A and B from the following road map reaction:



(1) C_6H_5OH, $C_6H_5BF_3$

(2) $C_6H_5N_2Cl$ and C_6H_5F

(3) $C_6H_5N_2Cl$ and C_6H_5Cl

(4) $C_6H_5N_2Cl$ and C_6H_5OH

Q34

The molar conductivity of acetic acid at infinite dilution is $390.7\ Scm^2\ mol^{-1}$. Conductivity of 0.1 M acetic acid solution is $5.2\ Scm^2\ mol^{-1}$, find out degree of dissociation of acetic acid :-

(1) 0.133

(2) 0.0133

(3) 0.139

(4) 0.001

Q35

What will be the correct decreasing order of sweetening capacity of sweetening agent given below:

I. Saccharin, II. Aspartame, III. Alitame.

(1) I > II > III

(2) III > II > I

(3) III > I > II

(4) II > I > III

Q36

CrO_4^{2-} (yellow) changes to $Cr_2O_7^{2-}$ (orange) in pH = x and vice - versa in pH= y. Hence, x and y may be-

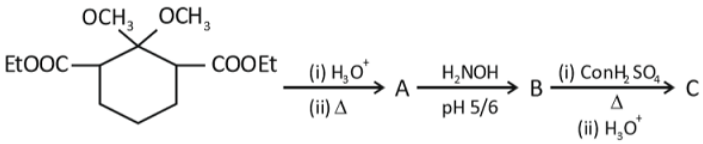
(1) 5, 9

(2) 6, 5

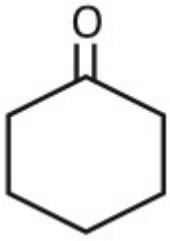
(3) 8, 6

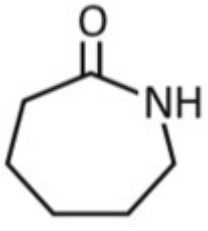
(4) 7, 7

Q37

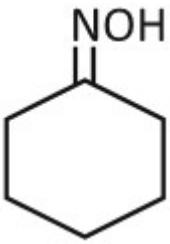


Compound C in above reaction sequence is

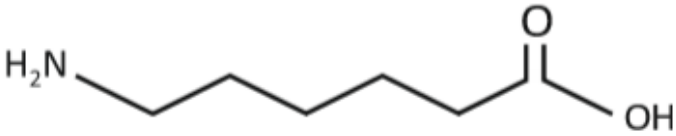
(1) 

(2) 

(3)



(4)



Q38

Which ion is responsible for the hardness in water?

(1) Ca^{+2}

(2) Na^{+}

(3) CI^{-}

(4) K

Q39

Which of the following statement is/are true -

(I) Borazine is aromatic

(II) There are four isotopic disubstituted borazine molecules $B_3N_3H_4X_2$

(III) Borazine is more reactive towards addition reactions than benzene

(IV) Banana bonds in B_2H_6 are longer but stronger than normal $B-H$ bonds

(1) I, II and III

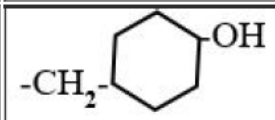
(2) I, II and IV

(3) I , II , III and IV

(4) Only II

Q40

Match the amino acid with its correct R group.

	Amino acid		R- group
P	Cysteine	I	$-CH_2OH$
Q	Serine	II	$-CH_2-CH_2-CH_2-COOH$
R	Glutamic acid	III	
S	Tyrosine	IV	$-CH_2-SH$

(1) P – I, Q – IV, R – II, S – III

(2) P – IV, Q – I, R – II, S – III

(3) P – III, Q – IV, R – II, S – I

(4) P – II, Q – I, R – IV, S – III

Q41

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Barium carbonate is insoluble in water and is highly stable.

Reason (R): The thermal stability of the carbonates increases with increasing cationic size.

Choose the most appropriate answer from the options given below :

(1) (A) is true but (R) is false.

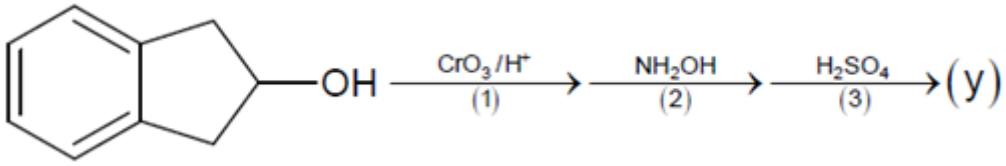
(2) Both (A) and (R) are true and (R) is the true explanation of (A).

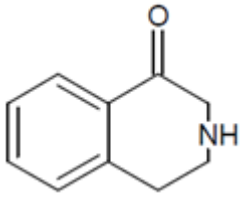
(3) (A) is false but (R) is true.

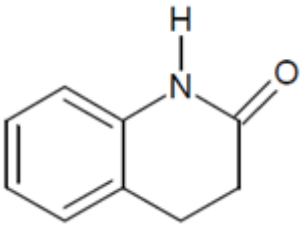
(4) Both (A) and (R) are true but (R) is not the true explanation of (A).

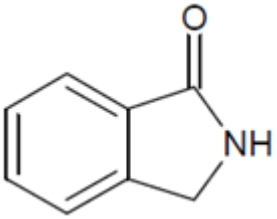
Q42

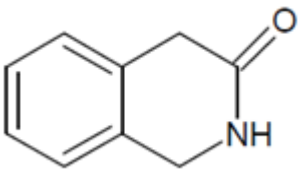
Major product of the following sequence of reactions could be :



(1)


(2)


(3)


(4)


Q43

Photochemical smog consists of excessive amount of X, in addition to aldehydes, ketones, peroxy acetyl nitrile (PAN). X is:

(1) CH_4

(2) CO_2

(3) O_3

(4) CO

Q44

In the Kjeldahl's method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.75 g of sample neutralized 10 mL of 1M H_2SO_4. The percentage of nitrogen in the soil is

(1) 37.33

(2) 43.33

(3) 45.33

(4) 35.33

Q45

Consider two reactions having same Arrhenius factor A, but different energies of activation.

(i) $A \rightarrow B;\ Ea_1 = 20\ kJ$

(ii) $C \rightarrow D;\ Ea_2 = 30\ kJ$

Both are at temperature $25^\circ C$

If temperature in both reaction is increased slightly in such a way that change in temperature in both case is same then, choose the correct option.

(1) The second reaction is faster

(2) The second reaction is more sensitive towards temperature variation

(3) If temperature increases, rate of first reaction increase more sharply

(4) All the above are correct

Q46

Which of the following involve froth flotation processes?

(I) Extraction of silver from argentite

(II) Extraction of iron from haematite

(III) Extraction of Cu from chalcopyrite

(IV) Extraction of Al from bauxite

(1) III and IV

(2) II and IV

(3) II and III

(4) I and III

Q47

Compounds (A) and (B) are –

$$Cl_2 \begin{cases} \xrightarrow{\text{Cold and dilute}} (A) + NaCl + H_2O \\ \xrightarrow{\text{Hot and conc}} (B) + NaCl + H_2O \end{cases}$$

(1) $NaClO_3, NaClO$

(2) $NaClO_2, NaOCl$

(3) $NaClO_4, NaClO_3$

(4) $NaOCl, NaClO_3$

Q48

Select the correct matching -

	List-I (Metal ions)		List- II Magnetic moment (BM)
(1)	XeF_4	(A)	Pyramidal
(2)	XeF_6	(B)	T-shape
(3)	XeO_3	(C)	Distorted octahedral
(4)	$XeOF_2$	(D)	Square planar

(1) $1-D, 2-C, 3-A, 4-B$

(2) $1-A, 2-B, 3-C, 4-D$

(3) $1-B, 2-A, 3-C, 4-D$

(4) $1-C, 2-A, 3-A, 4-B$

Q49

Assertion: Air quickly leaking out of a balloon becomes cooler.

Reason: The leaking air undergoes adiabatic expansion.

(1) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

(2) If both Assertion and Reason are true but Reason is not explanation of the Assertion.

(3) If Assertion is true but the Reason is false.

(4) If Assertion is false but Reason is true.

Q50

Match the following :

	Polymer		Monomer
i.	Nylon-6	a.	Ethene
ii.	Nylon-6,6	b.	Caprolactum
iii.	Polyethene	c.	Hexamethylene diamine and adipic acid

(1) (i)-(a),(ii)-(b),(iii)-(c)

(2) (i)-(a),(ii)-(c),(iii)-(b)

(3) (i)-(c),(ii)-(b),(iii)-(a)

(4) (i)-(b),(ii)-(c),(iii)-(a)

Q51

How many of the following features are $CORRECT$ regarding physisorption?

i. Highly specific in nature.

ii. Heat of adsorption is low.

iii. Favoured at low temperature.

iv. Reversible in nature.

v. Forms monomolecular layer of adsorbed particles.

Q52

The number of geometrical isomers possible for the complex $[\mathrm{Pt(NH_3)(NH_2OH)(Py)Cl}]^+$ is

Q53

How many of the following ethers CANNOT be prepared by Williamson's synthesis?

$\mathrm{CH_3OCH_2CH_3}, \mathrm{C_6H_5OCH_3}, (\mathrm{C_6H_5})_2\mathrm{O}, (\mathrm{CH_3})_3\mathrm{COCH_3}, (\mathrm{C_2H_5})_2\mathrm{O}, (\mathrm{CH_3})_3\mathrm{C-O-C(CH_3)_3}$

Q54

To a 25 mL $\mathrm{H_2O_2}$ solution, excess of acidified solution of potassium iodide was added. The iodine liberated required 20 mL of 0.3 N sodium thiosulphate solution. Calculate the volume strength of $\mathrm{H_2O_2}$ solution and report your answer by multiplying it with 1000.

Q55

For the reaction $\mathrm{N_2O_4} \rightleftharpoons 2\mathrm{NO_{2(g)}}$, the degree of dissociation of $\mathrm{N_2O_4}$ is 0.2 at $1\mathrm{atm}$. Then the $\mathrm{K_p}$ of $2\mathrm{NO_2} \rightleftharpoons \mathrm{N_2O_4}$ is

Q56

At what temperature (in kelvin) is the rms speed of hydrogen molecules the same as oxygen molecules at $1327°\mathrm{C}$?

Q57

A soft drink was bottled with a partial pressure of CO_2 of 3 bar over the liquid at room temperature. The partial pressure of CO_2 over the solution approaches a value of 30 bar when 44 g of CO_2 is dissolved in 1 kg of water at room temperature. The approximate pH of the soft drink is _____$\times 10^{-1}$.
(First dissociation constant of $H_2CO_3 = 4.0 \times 10^{-7}$; $\log 2 = 0.3$; density of the soft drink $= 1$ g mL^{-1})

Q58

In a face centred cubic arrangement of A and B atoms in which 'A' atoms are the corners of the unit cell and 'B' atoms are at the face centers. One of the 'A' atom is missing from one corner in unit cell. The simplest formula of compound is A_xB_y. The value of x is ____. (x and y are relative prime integers)

Q59

The work function (ϕ) of some metal is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength fall on the metal is

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
Ø(eV)	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

Q60

The decomposition reaction $2N_2O_5(g) \xrightarrow{\Delta} 2N_2O_4(g) + O_2(g)$ is started in a closed cylinder under the isothermal isochoric condition at an initial pressure of 1 atm. After $Y \times 10^3$ s , the pressure inside the cylinder is found to be 1.45 atm . If the rate constant of the reaction is $5 \times 10^{-4} s^{-1}$, assuming ideal gas behaviour, the value of 10 Y is

Q61

Let $f : R \to R$ be continuous function and $f(x) = f(2x)$ is true $\forall x \in R$ and $f(1) = 3$, then the value of $\int_{-1}^{1} f(f(x))dx$ is equal to:

(1) 0

(2) 2

(3) 6

(4) 12

Q62

Total number of solution of $\sin x \cdot \tan 4x = \cos x$ in $x \in (0, \pi)$ is/are:

(1) 7

(2) 6

(3) 5

(4) 4

Q63

Let m be a positive integer. If $\lim_{x \to 0} |\cos x + \sin 2x + \sin 3x|^{\cot x} = e^m$, then the value of m is:

(1) 2

(2) 3

(3) 4

(4) 5

Q64

Let $A = \left[a_{ij}\right]_{2\times 2}$ be a matrix where $a_{ij} \in \{2, 3\}$. If determinant of matrix A is non-negative, then probability that it is invertible is:

(1) $\frac{1}{2}$

(2) $\frac{5}{11}$

(3) $\frac{5}{16}$

(4) $\frac{3}{16}$

Q65

If complex number z satisfies $(z-\bar{z})^2 = 12|z|^2 - 4$ then find the maximum value of $3\sqrt{3}\,\mathrm{Re}(z) + 8\,\mathrm{Im}(z)$

(1) 5

(2) 7

(3) 9

(4) 6

Q66

If α and β are the distinct roots of the equation $\mathrm{x}^2 + (3)^{1/4}\mathrm{x} + 3^{1/2} = 0$, then the value of

$\alpha^{96}\left(\alpha^{12} - 1\right) + \beta^{96}\left(\beta^{12} - 1\right)$ is equal to :

(1) 56×3^{25}

(2) 56×3^{24}

(3) 52×3^{24}

(4) 28×3^{25}

Q67

Let $\vec{\mathbf{a}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}$ be three vectors of magnitude $2, 3, 5$ respectively, satisfying $|[\vec{\mathbf{a}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}]| = 30$

If $(2\vec{\mathbf{a}} + \vec{\mathbf{b}} + \vec{\mathbf{c}}) \cdot ((\vec{\mathbf{a}} \times \vec{\mathbf{c}}) \times (\vec{\mathbf{a}} - \vec{\mathbf{c}}) + \vec{\mathbf{b}}) = k$, then the value of $\left(\frac{k}{103}\right)$ is:

(1) 1

(2) 2

(3) 3

(4) 4

Q68

The minimum distance of the curve $y^2 = 2x^3 + 9 - 3x^2$ from point $Q(1, 0)$ is

(1) 2

(2) $2\sqrt{2}$

(3) $4\sqrt{2}$

(4) 8

Q69

Let matrix $\mathrm{A}=\begin{bmatrix}\mathrm{x} & \mathrm{y} & -\mathrm{z}\\ 1 & 2 & 3\\ 1 & 1 & 2\end{bmatrix}$, where $\mathrm{x},\ \mathrm{y},\ \mathrm{z}\in \mathrm{N}$. If $|\mathrm{adj}|\mathrm{adj}|\mathrm{adj}|\mathrm{adjA}||||=4^{8}\cdot 5^{16}$, then the number of such matrices A is equal to (where, $|M|$ represents determinant of a matrix M)

(1) 48

(2) 67

(3) 36

(4) 16

Q70

The locus of the centre of a circle which cuts the circle $\mathrm{x}^{2}-\ 20\mathrm{x}+\mathrm{y}^{2}+4=0$ orthogonally and also touches the line $\mathrm{x}=2$ is

(1) $\mathrm{y}^{2}=16\mathrm{x}+4$

(2) $\mathrm{x}^{2}=16\mathrm{y}$

(3) $\mathrm{x}^{2}\ =16\ \mathrm{y}+4$

(4) $\mathrm{y}^{2}=16\mathrm{x}$

Q71

If a & b are odd positive integers and c is even positive integer, then number of integral solutions of $a+b+c=16$, is -

(1) 28

(2) 21

(3) 36

(4) 45

Q72

Let a_k denotes the coefficient of x^k in the expansion of $(1+2x)^n, n \in N$.

If $\sum_{k=0}^{n}(3k+1)a_k = (px+q)r^n, p, q, r \in N$. Find $p+q+r$.

(1) 5

(2) 7

(3) 9

(4) 6

Q73

If the point $M(h,k)$ lie on the line $2x+3y=5$ such that $|MA-MB|$ is maximum where $A(2,3)$ and $B(1,2)$, then find the value of $(3h+2k)$

(1) 4

(2) 6

(3) 9

(4) 5

Q74

The value of θ for which the system of equations

$(\sin 3\theta)\mathrm{x} - 2\mathrm{y} + 3\mathrm{z} = 0$, $(\cos 2\theta\ )\mathrm{x} + 8\mathrm{y} - 7\mathrm{z} = 0$ and $2\mathrm{x} + 14\mathrm{y} - 11\mathrm{z} = 0$ has a non-trivial solution, is (here, $\mathrm{n} \in \mathrm{Z}$)

(1) $\mathrm{n}\pi$

(2) $\mathrm{n}\pi\ +\ (-1)^{\mathrm{n}}\ \pi/3$

(3) $\mathrm{n}\pi\ +\ (-1)^{\mathrm{n}}\ \pi/8$

(4) None of these

Q75

If the mean and the variance of the numbers $a,\ b,\ 8, 5$ and 10 are 6 and 6.8 respectively, then the value of a^3+b^3 is equal to

(1) 58

(2) 61

(3) 91

(4) 89

Q76

Area bounded between the curves $y=\sqrt{4-x^2}$ and $y^2=3|x|$ is :

(1) $\frac{\pi-1}{\sqrt{3}}$

(2) $\frac{2\pi-1}{3\sqrt{3}}$

(3) $\frac{2\pi-\sqrt{3}}{3}$

(4) none of these

Q77

Let the solution curve of the differential equation $xdy=\left(\sqrt{x^2+y^2}+y\right)dx, x>0$, intersect the line $\mathrm{x}=1$ at $y=0$ and the line $x=2$ at $y=\alpha$. Then the value of α is

(1) $\frac{1}{2}$

(2) $\frac{3}{2}$

(3) $-\frac{3}{2}$

(4) $\frac{5}{2}$

Q78

Let $\oplus$ and $\otimes$ are two mathematical operators. If $p\oplus\ (p\otimes q)$ is not a tautology, then $\oplus$ and $\otimes$ can be

(1) $\vee$ and $\Rightarrow$ respectively

(2) $\Rightarrow$ and $\wedge$ respectively

(3) $\Rightarrow$ and $\vee$ respectively

(4) None of these

Q79

The total number of divisors of the number $N = 2^5 \cdot 3^4 \cdot 5^{10} \cdot 7^6$ that are of the form $4K + 2,\ \forall K \in N$ is equal to

(1) 385

(2) 384

(3) 96

(4) 77

Q80

If $a + b + c = 6$ where $a, b, c \in R^+$then the maximum value of $a^3\ b^2 c$ is -

(1) $3^2 \cdot 2^3$

(2) $3^2 \cdot 2^2$

(3) $3^3 \cdot 2^2$

(4) 6

Q81

Let $I(a) = \int_0^\pi \ln\left(1 - 2a\cos x + a^2\right) dx$. Find the value of $\frac{I(100)}{I(10)}$.

Q82

If α is a real root of the equation $x^5 - x^3 + x - 2 = 0$, then find the value of $\left[\alpha^6\right]$.

[Note: $[\cdot]$ denotes the greatest integer function.]

Q83

When $(3053)^{456} - (2417)^{333}$ is divided by 9 the remainder is

Q84

Let $S_n = \sum_{x=1}^{n} x!; n \geq 6, T = \sin^{-1}\left(\sin\left(S_n - 7\left[\frac{S_n}{7}\right]\right)\right)$. If $\int_0^1 \frac{T}{\sqrt{1-x^2}}dx = \frac{a\pi}{b} - \pi^c$ where $a, b, c \in W$; (a and b are relatively prime integers) $b \neq 0$, then find $\left(\frac{b}{c} + a\right)$

Q85

Let $y^2 = x$ be a given parabola and a variable chord cuts the parabola at P and Q. Let C be the vertex of parabola. If the locus of the point of intersection of tangents at P and Q is $x + 1 = 0$, then the minimum area of triangle PCQ be M. Find $4M$

Q86

The value of the expression
$\tan\left(\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{2}{9}\right) + \tan^{-1}\left(\frac{1}{8}\right) + \tan^{-1}\left(\frac{2}{25}\right) + \tan^{-1}\left(\frac{1}{18}\right) + \ldots\ldots\infty\right)$, is:

Q87

Two variable complex numbers $\mathrm{z_1}$&$\mathrm{z_2}$ with their arguments in $(-\pi, \pi]$ are such that $|\mathrm{z_1} - (6 + 2\mathrm{i})| \leq 2\sqrt{2}$ and $\left|\mathrm{z_2} - (4\sqrt{2} + \mathrm{i}12\sqrt{2})\right| \leq \mathrm{k}$, where maximum value of $\arg(\mathrm{z_1})$ = minimum value of $\arg(\mathrm{z_2})$, then k is equal to

Q88

If $f(x) = \{x + \sin x\} + [x - \sin x] + [x]$ where $[y]$ and $\{y\}$ denote greatest integer function and fractional part function of y respectively, then find the number of points of discontinuity in $[0, \pi]$.

Q89

Let $\mathrm{L_1} : \frac{\mathrm{x}-6}{3} = \frac{\mathrm{y}-7}{-1} = \frac{\mathrm{z}-4}{1}$ and $\mathrm{L_2} : \frac{\mathrm{x}}{-3} = \frac{\mathrm{y}+9}{2} = \frac{\mathrm{z}-2}{4}$. If $\mathrm{L_1}$& $\mathrm{L_2}$ are skew lines such that P&Q are two points nearest to each other lying on lines $\mathrm{L_1}$& $\mathrm{L_2}$ respectively. If image of P with respect to plane $3x + 3y - z = 11$ is $R(a, b, c)$, then $(a + b + c)$ equals

Q90

If $[\vec{\mathbf{b}}\,\vec{\mathbf{c}}\,\vec{\mathbf{d}}] = 48$ and $(\vec{\mathbf{a}} \times \vec{\mathbf{b}}) \times (\vec{\mathbf{c}} \times \vec{\mathbf{d}}) + (\vec{\mathbf{a}} \times \vec{\mathbf{c}}) \times (\vec{\mathbf{d}} \times \vec{\mathbf{b}}) + (\vec{\mathbf{a}} \times \vec{\mathbf{d}}) \times (\vec{\mathbf{b}} \times \vec{\mathbf{c}}) + k\vec{\mathbf{a}} = \vec{0}$ (where $\vec{\mathbf{a}}$ is a non zero vector), then find value of k.

Answer Key

Q1 (1)	**Q2** (3)	**Q3** (1)	**Q4** (4)
Q5 (1)	**Q6** (2)	**Q7** (1)	**Q8** (3)
Q9 (1)	**Q10** (1)	**Q11** (1)	**Q12** (1)
Q13 (3)	**Q14** (2)	**Q15** (3)	**Q16** (1)
Q17 (4)	**Q18** (1)	**Q19** (2)	**Q20** (3)
Q21 (2)	**Q22** (3)	**Q23** (3)	**Q24** (2)
Q25 (1)	**Q26** (72)	**Q27** (75)	**Q28** (27)
Q29 (3)	**Q30** (24)	**Q31** (2)	**Q32** (3)
Q33 (2)	**Q34** (2)	**Q35** (3)	**Q36** (1)
Q37 (4)	**Q38** (1)	**Q39** (3)	**Q40** (2)
Q41 (2)	**Q42** (4)	**Q43** (3)	**Q44** (1)
Q45 (2)	**Q46** (4)	**Q47** (4)	**Q48** (1)
Q49 (1)	**Q50** (4)	**Q51** (3)	**Q52** (3)
Q53 (2)	**Q54** (1344)	**Q55** (6)	**Q56** (100)
Q57 (37)	**Q58** (7)	**Q59** (4)	**Q60** (23)
Q61 (3)	**Q62** (3)	**Q63** (4)	**Q64** (2)
Q65 (1)	**Q66** (3)	**Q67** (3)	**Q68** (2)

Q69 (3)	**Q70** (4)	**Q71** (1)	**Q72** (4)
Q73 (1)	**Q74** (1)	**Q75** (3)	**Q76** (3)
Q77 (2)	**Q78** (2)	**Q79** (2)	**Q80** (3)
Q81 (2)	**Q82** (3)	**Q83** (2)	**Q84** (6)
Q85 (4)	**Q86** (3)	**Q87** (8)	**Q88** (9)
Q89 (4)	**Q90** (96)		

Q1

For an isothermal process

$\mathrm{PV} = \text{constant} \ \therefore \ \mathrm{PV} = \mathrm{P_I 2V}$

$P_I = \frac{P}{2} \qquad \ldots(\mathrm{i})$

For an adiabatic process

$\mathrm{PV}^\gamma = \text{constant} \therefore \mathrm{PV}^\gamma = \mathrm{P_A}(2\mathrm{V})^\gamma$

or $P_A = \frac{P}{2^\gamma} \qquad \ldots(\mathrm{ii})$

Divide (i) by (ii), we get

$\frac{P_I}{P_A} = \frac{2^\gamma}{2} = 2^{\gamma-1}$

Q2

$\left(\frac{\mathrm{I_T}}{\mathrm{I_C}}\right)^2 = 1 + \frac{\mathrm{m}_a^2}{2}$

$I_T = 8.88\mathrm{A}, \ I_C = 8\mathrm{A}; \ \mathrm{m}_a = ?$

$\left(\frac{8.88}{8}\right)^2 = 1 + \frac{m_a^2}{2}$

$1.232 = 1 + \frac{m_a^2}{2}$

$\frac{m_a^2}{2} = 1.232 - 1 = 0.232$

$\mathrm{m_a} = \sqrt{2 \times 0.232} = 0.68 \text{ or } 68\%$

Q3

Material A is paramagnetic and Material B is ferromagnetic.

The susceptibility of material B is larger than A at given magnetic field because ferromagnetic material gets strongly magnetised and hence produces a larger intensity of magnetization in comparison to paramagnetic substance, therefore it is strongly magnetised.

Q4

From the definition of bulk modulus,

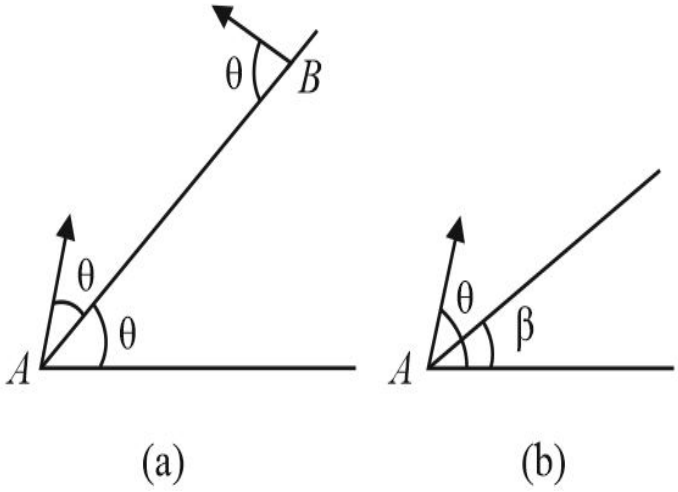
$\beta = -\frac{\Delta P}{\Delta V/V} = -V \cdot \frac{\Delta P}{\Delta V}$

Substituting the values, we have

$\beta = \frac{-(1.165-1.01)\times 10^5}{-10} \times 100 = 1.55 \times 10^5 \text{ Pa}$

Q5

Here, $\alpha = 2\theta$, $\beta = 0$



(a) (b)

Time of flight of $\mathrm{T_1} = \frac{2\mathrm{u}\sin(\alpha-\beta)}{\mathrm{g}\cos\beta}$

$= \frac{2\mathrm{u}\sin(2\theta-\theta)}{\mathrm{g}\cos\theta} = \frac{2\mathrm{u}}{\mathrm{g}}\tan\theta$

Time of flight of $\mathrm{T_2} = \frac{2\mathrm{u}\sin\theta}{\mathrm{g}\cos\theta} = \frac{2\mathrm{u}}{\mathrm{g}}\tan\theta$

So, $T_1 = T_2$. The acceleration of both the particles is g downwards. Therefore, relative acceleration between the two is zero or relative motion between the two is uniform. The relative velocity of A w.r.t. B is towards AB, therefore collision will take place between the two in mid air.

Q6

$\mathrm{T} = 20 \text{ N}$ and $\mathrm{kx} = 20 \text{ N}$

$\therefore\ \mathrm{x} = \frac{20}{40} = \frac{1}{2} \text{ m}$

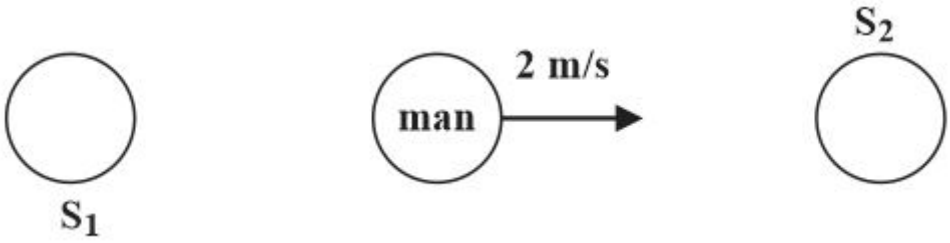
Now $\mathrm{mgx} = \frac{1}{2}\mathrm{mv}^2 + \frac{1}{2}\mathrm{kx}^2$

$\Rightarrow\ 5 \times 10 \times \frac{1}{2} = \frac{1}{2} \times 5\mathrm{v}^2 + \frac{1}{2} \times 40 \times \frac{1}{4}$

$\Rightarrow\quad 25 = 5 + \frac{5}{2}\mathrm{v}^2$

$\Rightarrow \mathrm{v} = 2\sqrt{2} \text{ m s}^{-1}$

Q7



$n_1 = n_0\left\{\frac{v-v_0}{v-v_s}\right\} = 800\left\{\frac{320-2}{320}\right\} = \frac{800\times318}{320}$ Hz

$n_2 = n_0\left\{\frac{v-v_0}{v-v_s}\right\} = 800\left\{\frac{320+2}{320}\right\} = \frac{800\times322}{320}$ Hz

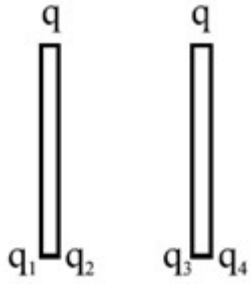
$\therefore$ Beat frequency = $n_2 - n_1$

$= \frac{800\,(322-318)}{320}$

$= \frac{800\times4}{320} = 10$ Hz

Q8

Capacitance does not depend on charge on it, $C = \frac{\varepsilon_0 A}{d}$



In final charge distribution, Charge on outer plates $q_1 = q_4 = q$ and inner plates $q_2 = q_3 = 0$

As charge on inner plates are zero so there is no field between the plates and no potential difference and no energy is stored.

Q9

According to the figure

$H = H_1 + H_2$

$$\frac{3\mathrm{KA}\left(100-\mathrm{T}\right)}{l}=\frac{2\mathrm{KA}\left(\mathrm{T}-50\right)}{l}+\frac{\mathrm{KA}\left(\mathrm{T}-0\right)}{l}$$

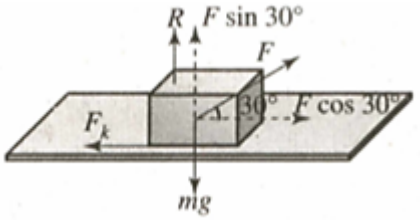
300 - 3T = 2T - 100 + T

6T = 400

Or $T=\frac{200}{3}{}^{o}\mathrm{C}$

Q10

FBD of the block is shown below



Kinetic friction acting on the body$=\mu_k R=0\ .2(Mg-F\sin 30^{\circ})$

$=0\ .2\left(5\times 10-40\times\frac{1}{2}\right)=0\ .2(50-20)=6\ \mathrm{N}$

By Newton's second law of motion, acceleration of the block

$=\frac{net\ force}{mass}=\frac{F\cos 30^{\circ}-kinetic\ friction}{mass}=\frac{40\times\frac{\sqrt{3}}{2}-6}{5}=5\ .73\ \ \mathrm{m/s^2}$

Q11

The equation for the given v - x graph is

$v=-\frac{v_0}{x_0}x+v_0$ ...(i)

Differentiating the above equation w.r.t. x we get

$\frac{dv}{dx}=-\frac{v_0}{x_0}$

Multiplying both sides of the above equation by v, we get

$v\frac{dv}{dx}=-\frac{v_0}{x_0}\times v=-\frac{v_0}{x_0}\left[-\frac{v_0}{x_0}x+v_0\right]$ From (i),

$\therefore a=\frac{v_0^2}{x_0^2}x-\frac{v_0^2}{x_0}$ ...(ii) $\left[\because a=v\frac{dv}{dx}\right]$

On comparing equation (ii) with equation of a straight line $y = mx + c$ where m is the slope of the line and c is its intercept on y-axis we get, $m = \frac{v_0^2}{x_0^2}$ [therefore, m will always be positive]

$\Rightarrow$ The slope m or $\tan\theta = 0$ or $'\theta'$ is an acute angle.

Also, the comparison gives -

$c = \frac{-v_0^2}{x_0}$ or the intercept will be negative.

These conditions are met only in this graph.

Q12

For X: energy $E_1 = 200 \times 7.4 = 1480MeV$

For A: energy $E_2 = 110 \times 8.2 = 902MeV$

For B: energy $E_3 = 80 \times 8.1 = 648MeV$

Energy Released E = -$\mathrm{E}_1 + E_2 + E_3$

$E = (902 + 648) - 1480 = 70MeV$

Q13

Let $\mathrm{M} \propto \left[\mathrm{F}^{\mathrm{a}}\mathrm{L}^{\mathrm{b}}\mathrm{T}^{\mathrm{c}}\right]$

Writing dimensions on both sides and using the principle of homogeneity of dimensions we have,

$\left[\mathrm{M}^1\mathrm{L}^0\mathrm{T}^0\right] = \mathrm{k}\left[\mathrm{M}\,\mathrm{L}\,\mathrm{T}^{-2}\right]^{\mathrm{a}}[\mathrm{L}]^{\mathrm{b}}[\mathrm{T}]^{\mathrm{c}}$

On comparing the power on both sides we get $a = 1, a + b = 0$ and $-2a + c = 0$

On solving we have $b = -1, c = 2, a = 1$

$\therefore$ units of mass is $\left[\mathrm{F}\,\mathrm{L}^{-1}\mathrm{T}^2\right]$

Q14

In figure A , two springs are connected in parallel. The effective spring constant is, $\mathrm{k}_{\mathrm{eff}} = \mathrm{k}_1 + \mathrm{k}_2$

$\therefore \mathrm{T} = 2\pi\sqrt{\frac{\mathrm{m}}{\mathrm{k}_1 + \mathrm{k}_2}}; \mathrm{A} - \mathrm{s}$

In figure B, two identical spring are connected in parallel. The effective spring constant is

$\mathrm{k}_{\mathrm{eff}} = \mathrm{k} + \mathrm{k} = 2\mathrm{k} \therefore \mathrm{T} = 2\pi\sqrt{\frac{\mathrm{m}}{2\mathrm{k}}}; \mathrm{B} - \mathrm{r}$

In figure C, two identical spring are connected in series. The effective spring constant is

$\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{k_2+k_1}{k_1 k_2} \text{or } k_{eff} = \frac{k_1 k_2}{k_1+k_2}$

$\therefore T = 2\pi\sqrt{\frac{m(k_1+k_2)}{k_1 k_2}}; C - p$

In figure D, two identical spring are connected in series. The effective spring constant is

$k_{eff} = \frac{(k)(k)}{k+k} = \frac{k}{2} \therefore T = 2\pi\sqrt{\frac{2m}{k}}; D - q$

Q15

Assertion is correct because $EMF \propto \cos\theta$

Reason is wrong because field is steady.

Induced EMF in AC generator is given by $E = BAN\omega\cos(\omega t)$

where $B -$ *uniform magnetic field strength*

$A -$ *area of the coil*

$N -$ *number of turns of the coil*

$\omega -$ *angular velocity of the rotating coil*

$t -$ *instantaneous time*

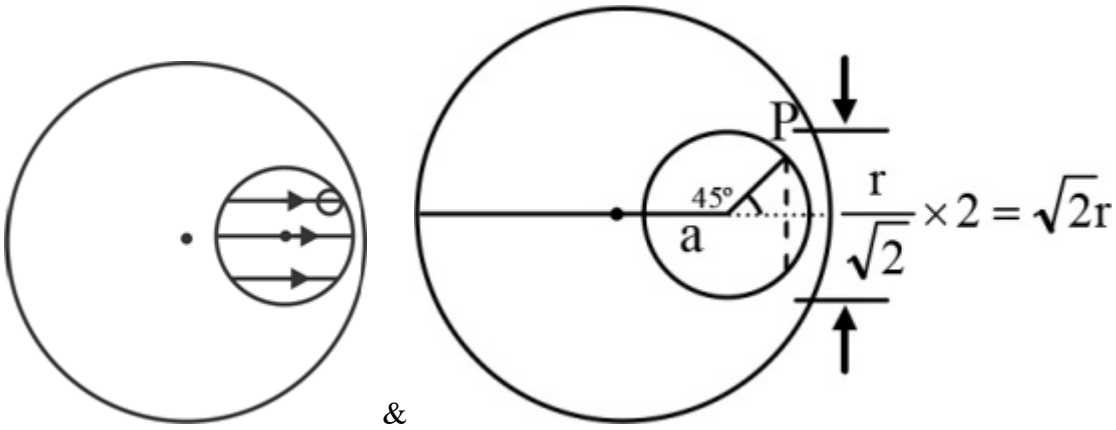
$\theta = \omega t,$ *angle between plane of the coil and magnetic field*

If θ *is zero, induced emf is maximum*

Q16

$E = \frac{\rho a}{3\varepsilon_0} \; F = eE = \frac{e\rho a}{3m\varepsilon_0}$

$\text{Acceleration} = \frac{\rho e a}{3m\varepsilon_0}$



&

$= \frac{1}{2} \times \frac{\rho e a}{3m\varepsilon_0} \times t^2 = \sqrt{2}r$

Electron will move opposite to electronic field with constant accleration along shown horizontal line of length $= \sqrt{2}r$

Q17

If the loop current is coming out it should be taken as positive and if it is going in it is taken as negative.

i_3 is coming out and again going in so its net contribution will be zero.

According to Ampere's law,

$$\oint \vec{B} \cdot \vec{dl} = \mu_0(\Sigma i_{inside})$$
$$= \mu_0(i_1 + i_2 + i_3 - i_3)$$
$$= \mu_0(i_1 + i_2)$$

Q18

In parallel combination of electrical appliances,

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

As voltage (V) across each appliance is same, so

$$\frac{V^2}{R} = \frac{V^2}{R_1} + \frac{V^2}{R_2} + \frac{V^2}{R_3}$$
$$P = P_1 + P_2 + P_3$$

Thus, both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

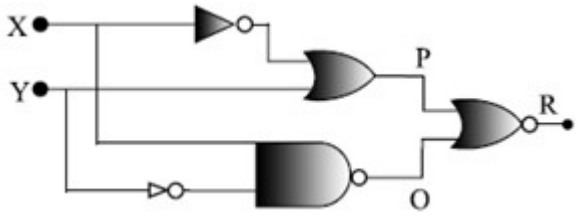
Q19

The oil drop experiment was to measure the specific charge of an electron.

Davisson & Germer experiment proved the wave nature of electrons.

Rutherford experiment proved the presence of a nucleus, while the Franck-Hertz experiment established the Quantisation of energy levels.

Q20



From figure

$$\left[\overline{\left(\overline{X}+Y\right)+X\overline{Y}}\right]=R$$

$$R=\left(\overline{\overline{X}+Y}\right)\cdot\left(\overline{X\overline{Y}}\right)$$

$$R=\left(\overline{X}\,\overline{Y}\right)\left(\overline{X}+\overline{Y}\right)$$

$$R=X\overline{Y}\left(\overline{X}+Y\right)$$

$$R=X\overline{X}\,\overline{Y}+X\overline{Y}$$

$$R=X\overline{Y}$$

i.e., when $\left.\begin{matrix}X=1\\Y=0\end{matrix}\right]\Rightarrow R=1$

Q21

Average K.E. per molecule of a gas at the given temperature (T) is given as,

$\frac{1}{2}mv^2=\frac{3}{2}kT$

where, m= molecular mass of the gas

$\therefore\ \ mv=\sqrt{3mkT}$

But, $\lambda=\frac{h}{mv}=\frac{h}{\sqrt{3mkT}}$

$\therefore\ \ \frac{\lambda_{11}}{\lambda_{11}}=\sqrt{\frac{m_{H_1}T_{H_2}}{m_{11}T_{11}}}=\sqrt{\frac{(4)(273+327)}{(2)(273+27)}}=\sqrt{4}=2$

Q22

From lens maker's equation,

$$P = \frac{1}{f} = \left(\mu - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

For plano - convex lens,

$$\mathrm{R}_1 = \infty, \mathrm{R}_2 = -\mathrm{R}$$

$$\therefore \quad P_1 = (\mu_1 - 1)\left(\frac{1}{\infty} - \frac{1}{-R}\right) = \frac{(\mu_1 - 1)}{R}$$

For plano-concave lens,

$$\mathrm{R}_1 = -\mathrm{R}, \mathrm{R}_2 = \infty$$

$$\therefore \quad \mathrm{P}_2 = (\mu_2 - 1)\left(\frac{1}{-\mathrm{R}} - \frac{1}{\infty}\right) = \frac{-(\mu_2 - 1)}{\mathrm{R}}$$

Given that,

$$\mathrm{P}_2 = 2\mathrm{P}_1$$

$$\therefore \quad \frac{-(\mu_2 - 1)}{\mathrm{R}} = \frac{2(\mu_1 - 1)}{\mathrm{R}}$$

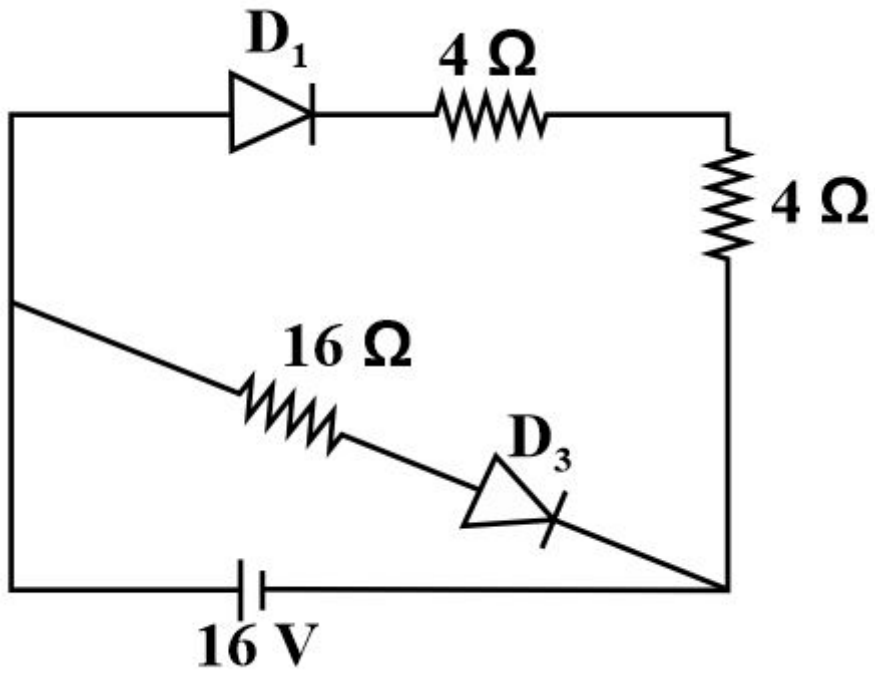
$$\therefore \quad -\mu_2 + 1 = 2\mu_1 - 2$$

$$\therefore \quad 2\mu_1 + \mu_2 = 3$$

$$\therefore \quad \mu_1 + \frac{\mu_2}{2} = 1.5$$

Q23

In the given circuit diode D_2 is reverse biased, so it will not conduct. Diode D_1 and D_3 are forward biased, so they will conduct, Thus the equivalent circuit is



$\therefore \mathrm{R_{net}} = \frac{(4+4)\times 16}{(4+4)+16} = \frac{8\times 16}{24} = \frac{16}{3}\ \Omega$

$\therefore$ current through the battery,

$\mathrm{I} = \frac{\mathrm{V}}{\mathrm{R_{net}}} = \frac{16}{\frac{16}{3}} = 3\ \mathrm{A}$

Q24

$\mathrm{K_f} - \mathrm{K_i} = \int \mathrm{P}\cdot \mathrm{dt}$

$\frac{1}{2}\mathrm{mv}^2 - 0 = \int_0^2 \left(\frac{3}{2}\mathrm{t}^2\right)\mathrm{dt}$

$\frac{1}{2}\left(2\right)\mathrm{v}^2 = \frac{3}{2}\left[\frac{\mathrm{t}^3}{3}\right]_0^2 = 4$

$\mathrm{v} = 2\ \mathrm{m\ s^{-1}}$

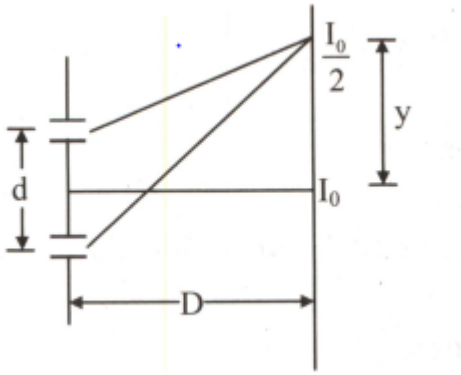
Q25

First Recall the formula of power factor, $\cos\phi = \frac{R}{Z}$ $\Rightarrow \cos\phi_{\text{initial}} = \frac{R}{Z} = \frac{R}{\sqrt{(X_{\mathrm{L}})^2+R^2}} = \frac{R}{\sqrt{9R^2+R^2}} = \frac{1}{\sqrt{10}}.$

$\Rightarrow \cos\phi_{final} = \frac{R}{Z} = \frac{R}{\sqrt{(X_{\mathrm{L}}-X_C)^2+R^2}} = \frac{R}{\sqrt{R^2+R^2}} = \frac{1}{\sqrt{2}}$

$\Rightarrow \frac{\cos\phi_{final}}{\cos\phi_{\text{initial}}} = \frac{\sqrt{5}}{x} \Rightarrow x = 1$

Q26



$I = I_0 \cos^2 \frac{\phi}{2}$

$\Rightarrow \frac{I_0}{2} = I_0 \cos^2 \frac{\phi}{2}$

$\because \cos \frac{\phi}{2} = \frac{1}{\sqrt{2}}$

$\therefore \quad \frac{\phi}{2} = \frac{\pi}{4}$

$\therefore \quad \phi = \frac{\pi}{2}$

since $\frac{2\pi}{\lambda}\left(\frac{yd}{D}\right) = \frac{\pi}{2} \quad \ldots . \left(\text{Path difference } = \frac{yd}{D}\right)$

$\therefore \quad y = \frac{\lambda D}{4d}$

$= \frac{6000 \times 10^{-10} \times 120 \times 10^{-2}}{4 \times 0.25 \times 10^{-2}}$

$= 7.2 \times 10^{-5} \mathrm{m}$

$= 72 \mu \mathrm{m}$

Q27

de-Broglie wavelength is

$\lambda = \frac{\mathrm{h}}{\mathrm{p}} = \frac{\mathrm{h}}{\sqrt{2\,\mathrm{mE}}}$

When K.E. is increased by 15 times,

$\mathbf{E'} = \mathbf{E} + 15\mathbf{E} = 16\mathbf{E}$

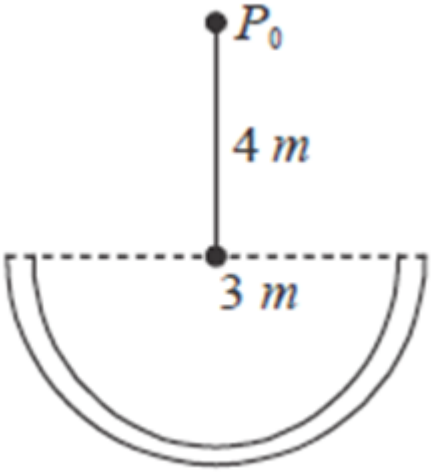
$\therefore \ \lambda' = \frac{\mathrm{h}}{\sqrt{2\mathrm{m}(16\mathrm{E})}} = \frac{1}{4}\frac{\mathrm{h}}{\sqrt{2\,\mathrm{mE}}} = \frac{\lambda}{4} = 0.25\lambda$

$\therefore \ \%change = \lambda - \lambda' = \lambda - 0.25\lambda$

$= 0.75\lambda \Rightarrow 75\%$

Note: There is a word increased by and not increased to. Most students get confused by the language of question. Similar thing is given in PYQs as well

Q28



Total power emitted by shell due to radiation

$P_{\text{out}} = \sigma\left(2\pi R^2 + \pi R^2\right)T^4 = 3\pi R^2\sigma T^4$

Power gained by the shell from P_0 :

$P_{\text{in}} = P_0\frac{2\pi r^2\left(1-\cos 37^\circ\right)}{4\pi r^2} = \frac{P_0}{10}$

Equating: $\frac{P_0}{10} = 3\pi R^2\sigma T^4$

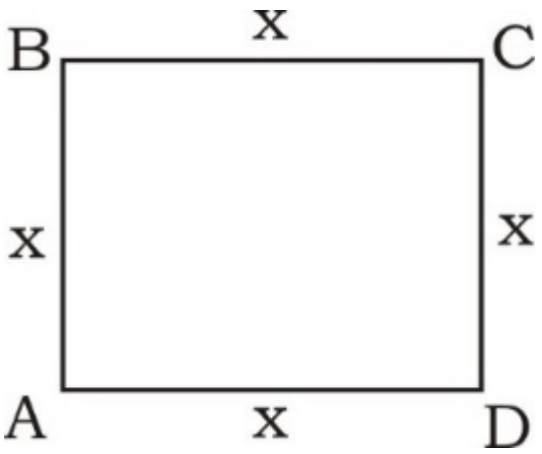
Gives $T^4 = \frac{P_0}{270\pi\sigma}$

$$n = \frac{270}{10} = 27$$

Q29

Let the equivalent resistance of one infinite ladder be x. Then the complete network reduces to

$\therefore\ R_{\mathrm{AB}} = \frac{x \times 3x}{x+3x} = \frac{3}{4}x$ and $R_{\mathrm{AC}} = \frac{2x \times 2x}{2x+2x} = x$;

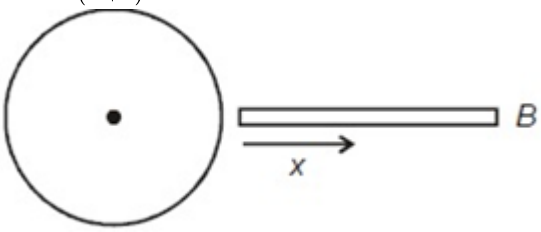


Hence $\frac{R_{\mathrm{AB}}}{R_{\mathrm{AC}}} = \frac{3}{4}$

Q30

Net force on the rod is equal to centripetal force.

$f_g = \frac{GMm}{(R+x)^2 R}dx$



$= \frac{GMm}{R}\int_0^R \frac{dx}{(R+x)^2}$

$f_g = \frac{GMm}{2R^2}$

Q31

$\pi_1 V_1 + \pi_2 V_2 = \pi_R (V_1 + V_2)$

$1 \times 1 + 3.5V = 2.5(1 + V)$

$1 + 3.5V = 2.5 + 2.5V$

or $V = 1.5$ L

Q32

(I) $CH_3 - \overset{\cdot\cdot}{\underset{\cdot\cdot}{O}} - \overset{\oplus}{C}H_2$is more stable than $CH_3 - \overset{\oplus}{C}H_2$, becausse of +M-effect of $CH_3 - \overset{\cdot\cdot}{\underset{\cdot\cdot}{O}}-$.

(II)$Me_3\overset{\oplus}{C}H$ is more stable than $CH_3 - CH_2 - \overset{\oplus}{C}H_2$,

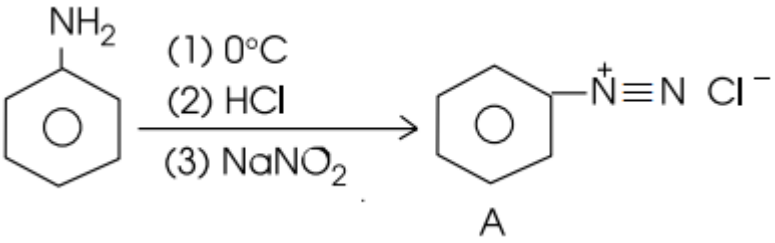
because $Me_3C^{\oplus}$ have more 2-hydrogen atoms than $CH_3 - CH_2 - CH_2^{\oplus}$, so, it is more stable, due to greater hyperconjugation.

(III) The positive charge is delocalised in $CH_2 = CH - \overset{\oplus}{C}H_2$, so more stable than,$CH_3 - CH_2 - CH_2^{\oplus}$.

(IV) $CH_2 = \overset{\oplus}{C}H$ will be less stable than $CH_3 - \overset{\oplus}{C}H_2$ as positive charge is less stable on double bonded 'C' than on single bonded 'C'.

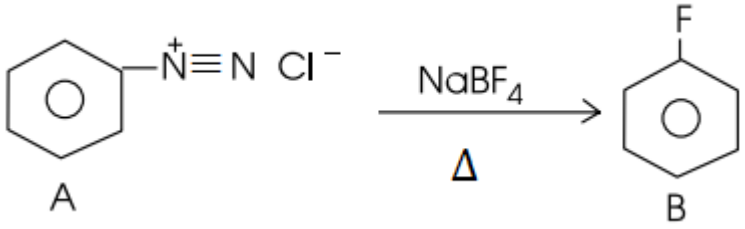
Q33

Benzenediazonium chloride is prepared by the reaction of aniline with nitrous acid at $273 - 278$ K. Nitrous acid is produced in the reaction mixture by the reaction of sodium nitrite with hydrochloric acid.



A

When benzene diazonium chloride is treated with sodium fluoroborate, benzene diazonium fluoroborate is precipitated which on heating decomposes to yield benzyl fluoride.



Q34

$\alpha = \frac{\Lambda_m}{\Lambda_m^\infty} = \frac{5.2}{390.7} = 0.0133$

Q35

This problem includes conceptual mixing of sweetening agent and their sweetening capacity

Sweetening agent: The substances other than sugar, used for sweetening of food and enhancing odour and flavour are known as sweetening agents. Commonly used sweetening agents are Saccharin, Aspartame, Alitame etc.

Comparative study of sweetening agent:

Sweetening capacity of saccharin = $500\times$ sweetening capacity of sugar

Sweetening capacity of aspartame = $180\times$ sweetening capacity of sugar

Sweetening capacity of alitame = $2000\times$ sweetening capacity of sugar

Hence, the correct order of sweetening capacity is represented by choice (III > I > II).

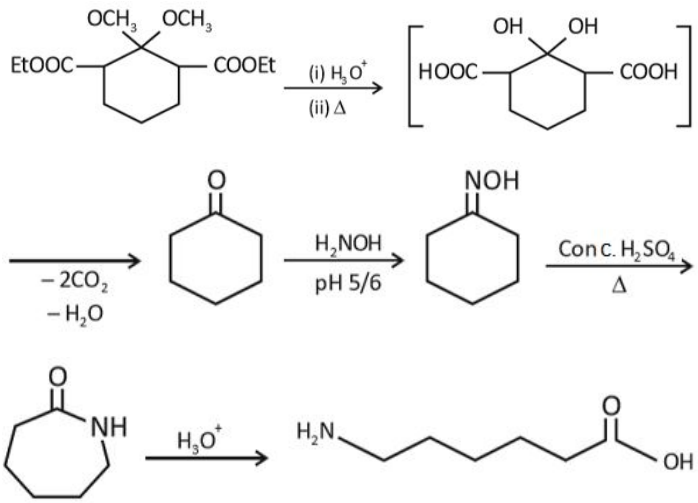
Q36

$2CrO_4^{2-} + 2H^+ \longrightarrow Cr_2O_7^{2-} + H_2O$ [where x= pH = 4 - 5]

$Cr_2O_7^{2-} + 2OH^- \longrightarrow 2CrO_4^{2-} + H_2O$ [where y =pH = 9 - 10]

so pH value is 5 and 9 respectively

Q37

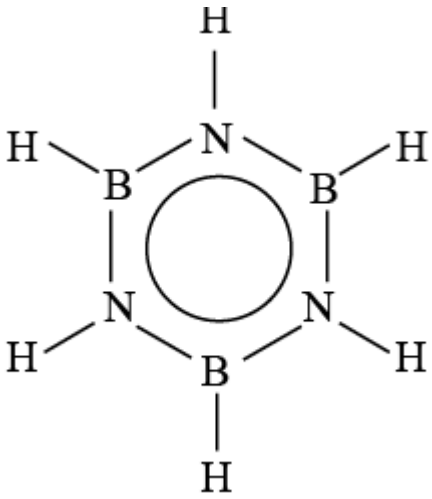


Q38

The ions that cause hardness in water are Ca^{+2} and Mg^{+2}. They form salts with HCO_3, Cl^-, $SO_4{}^{-2}$ions which when dissolved in water, cause the hardness of the water.

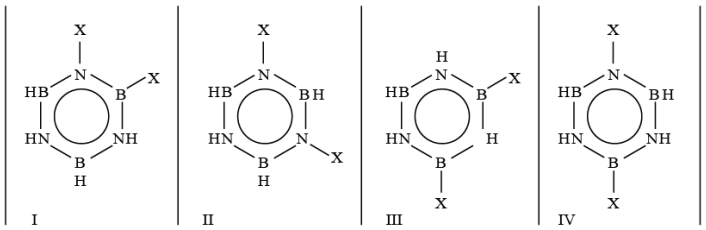
Q39

I. Borazine is aromatic - True



Each Nitrogen donates two electrons of its lone pair to the pi electron cloud. Hence with 6π electron, it follows Huckels Rule of anomorich $(4n+2)\pi$ electron in ring.

II. There are four isomeric distributed borazine molecules $B_3\ N_3H_4X_2$ - True.

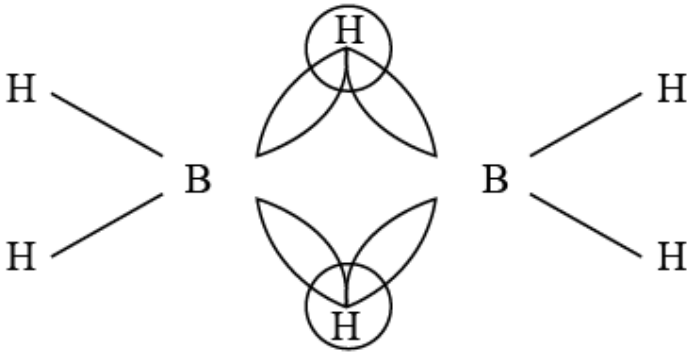


4 possible structures

III. Bonazine is more reactive towards addition reactions than benzene.

Benzene (C_6H_6) contains $C-C$ bonds of a π saturated system. Where Bonazine contains $B-N$ bond with electronegativity difference which makes the π-electron cloud to be more localized on N atoms rather the B. This makes bonozire more prone to addition reactions.

IV. Banana bonds in. B_2H_6 are longer but stronger than normal B-H bonds - True



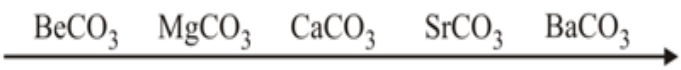
Banana bonds are longer due to the $3c-2e^-$involvement. But they are stronger than normal $B-H$ bonds.

Option C is correct - I, II, III, IV

Q40

Amino acids are monomers of proteins. These are substituted methanes and has a variable group R attached to the carbon. Due to presence of R group various types of amino acids are formed. Cysteine and Methionine are Sulphur containing amino acids. Tyrosine are amino acids with benzene ring. Glutamic acid is acidic and that means have acidic group. Serine has alcoholic group.

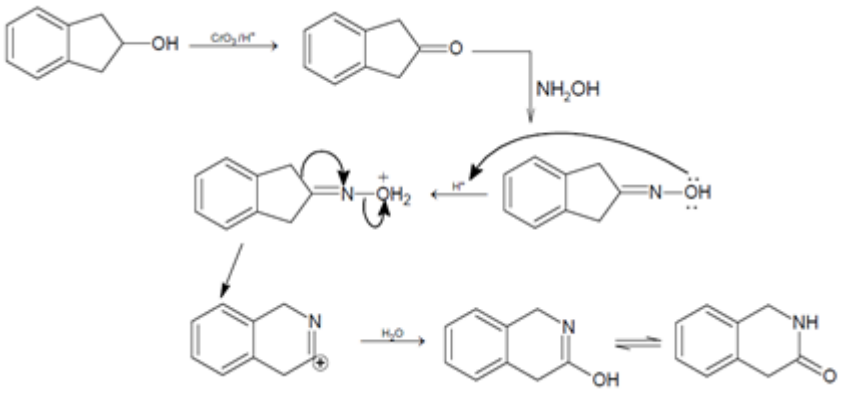
Q41



(i) Stability ↑ as size of cation ↑, ionic character ↑

(ii) Solubility in water ↓

Q42



Q43

Photochemical smog is formed on the chemical reaction of sunlight, nitrogen oxides and volatile organic compounds in the atmosphere.

$\mathrm{NO_2(g)} \xrightarrow{\mathrm{hv}} \mathrm{NO(g)+O(g)}$

Oxygen atom (nascent) is very reactive and combines with $\mathrm{O_2}$ in air to produce ozone.

$\mathrm{O(g)+O_2(g)} \rightarrow \mathrm{O_3(g)}$

Q44

$\because \mathrm{M} \times \mathrm{V(ml)} = \mathrm{m\ mol}$

$10\ \mathrm{m\ mol\ H_2SO_4} = 20\ \mathrm{m\ mol\ of\ NH_3}$

$[\mathrm{H_2SO_4 + 2NH_3 \rightarrow (NH_4)_2SO_4}]$

$1\ \mathrm{mol\ NH_3}$ contains 14 g nitrogen

$20 \times 10^{-3}\ \mathrm{mol\ NH_3}$ contains $14 \times 20 \times 10^{-3}$ nitrogen

0.75 g of sample contains

$\%\ \text{Nitrogen} = \frac{14 \times 20 \times 10^{-3}}{0.75} \times 100 = 37.33\%$

Q45

Arrhenius equation;

$$d(\ln k)=\frac{E_a}{RT^2}dT$$

$$\frac{d(\ln k)}{dT}=\frac{E_a}{RT^2}$$

Hence, when $dT=$ same, the change in $d(\ln K)$ wil be highest when E_a is highest.

So, if temperature increases, then rate of reaction having more E_a increases sharply and the reaction becomes more sensitive to temperature changes.

Q46

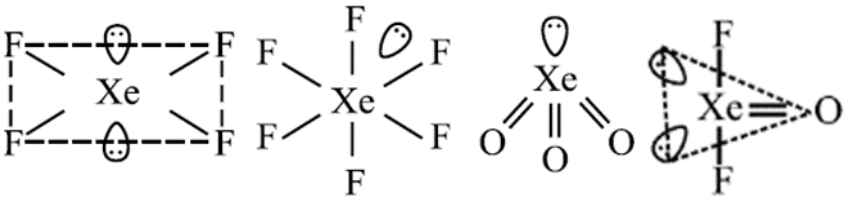
Ag_2S and $CuFeS_2$ are sulphide ores and froth flotation is used for the concentration of the ore.

Q47

$$Cl_2+NaOH\xrightarrow{\text{Cold\&dil}}NaCl+NaOCl$$

$$Cl_2+NaOH\xrightarrow{\text{Hot\&Conc.}}NaCl+NaOCl_3$$

Q48



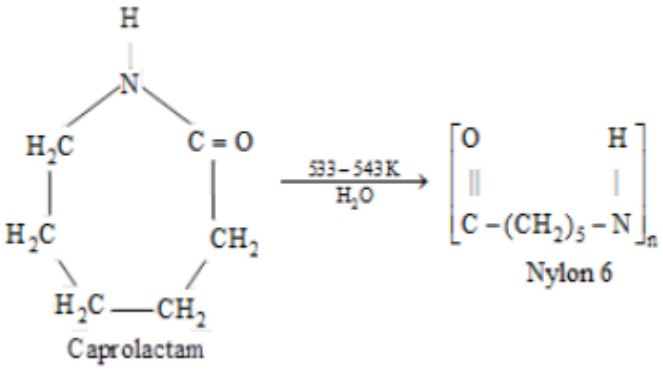
Hence the correct match is $1-D, 2-C, 3-A, 4-B$

Q49

When a system undergoes a change under the condition that there is no exchange of heat between the system and surroundings, then the process is called an adiabatic process. The leaking air of balloon undergoes adiabatic expansion. Air cools down due to adiabatic expansion as air has to do work against the external pressure at the cost of its internal energy. Due to work done against external pressure, the internal energy of air reduces. Thus, it becomes cooler.

Q50

(i) Nylon-6



(ii) Nylon-6, 6

$$n\,H_2N(CH_2)_6NH_2 + n\,HOOC(CH_2)_4COOH \rightarrow \underset{\text{Nylon 6, 6}}{-[NH(CH_2)_6NHCO(NH_2)_4CO]_n-} + nH_2O$$

(iii) Polythene

$$n\,CH_2 = CH_2 \xrightarrow{\text{Polymerisation}} -[CH_2 - CH_2]_n-$$

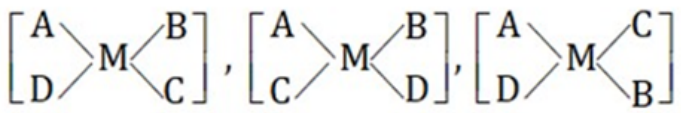
Ethene Polythene

Q51

Points (ii), (iii) and (iv) are correct regarding physisorption.

Q52

Complex is square planar



Q53

In Williamson's synthesis,

i. 1° alkyl halide and 3° alkoxide give best yield of corresponding ethers.

ii. In case of, 3° alkyl halide, elimination competes over substitution and alkene is obtained as a product.

iii. Aryl halides do not undergo nucleophilic substitution unless it is activated by the presence of electron withdrawing group. Thus, $(C_6H_5)_2O$ and $(CH_3)_3C-O-C(CH_3)_3$ cannot be prepared by Williamson's synthesis.

Q54

Meq of H_2O_2 = Meq of I_2 = Meq of $Na_2S_2O_3$.

If N is normality of H_2O_2, then

N x 25 = 0.3 x 20, $N_{H_2O_2} = 0.24N$

$M_{H_2O_2} = 0.12$ M

N = 0.24

Volume strength = 0.12×11.2

=1.344 Vol

Final answer $= 1.344 \times 1000 = 1344$

Q55

	$N_2O_4(g) \rightleftharpoons$	$2NO_2(g)$
at t = 0	1	–
at t = 1	$1-\alpha$	2α
mole fraction	$\frac{1-\alpha}{1+\alpha}$	$\frac{2\alpha}{1+\alpha}$

$K_p = \frac{[P_{NO_2}]^2}{P_{N_2O_4}}$

$K_p = \frac{4\alpha^2}{1-\alpha^2}p$

$\Rightarrow K_p = \frac{4\times(0.2)^2}{1-(0.2)^2} = \frac{0.16}{0.96}$

for $2NO_2 \rightleftharpoons N_2O_4$; $K_p = \frac{0.96}{0.16} = 6$

Q56

$\mathrm{v_{H2}} = \mathrm{v_{O2}}$

$\sqrt{\frac{3\mathrm{RT_{H2}}}{\mathrm{MH_2}}} = \sqrt{\frac{3\mathrm{RT_{O2}}}{\mathrm{MO_2}}}$

So, $\sqrt{\mathrm{M_{O2}T_{H2}}} = \sqrt{\mathrm{M_{H2}T_{O2}}}$

$32 \times \mathrm{T_{H2}} = 2 \times 1600$

$\mathrm{T_{H2}} = 2 \times \frac{1600}{32} = 100$

Q57

Amount of CO_2 in one liter of solution $= 4.4 . = 0.1$ Mole

$\mathrm{pH} = 1/2\{\mathrm{pKa} - \mathrm{logC}\}$

For a weak acid solution

$\mathrm{pH} = 1/2\{6.4 + 1\} = 3.7$

Q58

Number of atoms of A at corners $= 7$ (one 'A' is missing)

$\therefore$Contribution of atoms of 'A' to the unit cell

$= 7 \times \frac{1}{8} = \frac{7}{8}$

Total number of atoms 'B' at face $= 6$

$\therefore$ Contribution of atom 'B' to the unit cell

$= 6 \times \frac{1}{2} = 3$

$A : B = \frac{7}{8} : 3 = 7 : 24$

Hence, formula $= A_7B_{24}$

$\therefore$The value of x is 7

Q59

Energy of light $= \frac{1240}{300} = 4.13\text{eV}$, so the metal having work function lower than 4.13eV will emit photo-electron, so the following 4 metals will show photoelectric effect

Metal	Li	Na	K	Mg
Ø(eV)	2.4	2.3	2.2	3.7

Note: For photoelectric effect to occur, the energy of a photon must be greater than the work function of the metal is the explanation

Q60

	$2N_2O_5 \xrightarrow{\Delta} 2NO_2(g) + O_2(g)$		
t = 0	1 atm	0	0
t = t	(1-P)	P	$\frac{P}{2}$
t = ∞	0	1 atm	0.5 atm

$P_0 = 1 \text{ atm}, \ P_t = 1.45 \ \text{ atm}, P_\infty = 1.5 \ \text{ atm}$

From the unit of $K = 5 \times 10^{-4}s^{-1}$,

The reaction is first-order

$t = \frac{2.303}{2\times5\times10^{-1}} \log\left(\frac{1.5-1}{1.5-1.45}\right)$

$\mathrm{t} = 2\ .303 \times 10^3 = \mathrm{y} \times 10^3$

$\therefore y = 2.303$

After round off: $\mathrm{y} = 2\ .3$

Q61

$\because \quad f(2x) = f(x)$ replace x by $\frac{x}{2}$

$\therefore \quad f(x) = f\left(\frac{x}{2}\right) = f\left(\frac{x}{2^2}\right) = \ldots\ldots\ldots = f\left(\frac{x}{2^n}\right)$

$\Rightarrow \quad \lim_{n\to\infty} f(x) = \lim_{n\to\infty} f\left(\frac{x}{2^n}\right) \Rightarrow f(x) = f(0) = \text{ constant}$

$\because \quad f(1) = 3$

$\therefore \quad f(x) = 3 \forall x \in R$

$\therefore \quad \int_{-1}^{1} f(f(x))dx = \int_{-1}^{1} 3dx = 6.$

Q62

$$\sin x \cdot \tan 4x = \cos x$$
$$\sin x \cdot \sin 4x = \cos x \cdot \cos 4x$$
$$\cos 4x \cdot \cos x - \sin x \cdot \sin 4x = 0$$
$$\cos(4x + x) = 0$$
$$\cos 5x = 0$$
$$5x = (2n-1)\frac{\pi}{2} \quad (n \in I)$$
$$x = (2n-1)\frac{\pi}{10}$$
$$\therefore \quad x = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}$$

Q63

$$\lim_{x\to 0} |\cos x + \sin 2x + \sin 3x|^{\cot x} = e^m$$
$$= e^{\lim_{x\to 0}(|\cos x + \sin 2x + \sin 3x| - 1)\cot x} = e^m$$
$$\Rightarrow \quad m = \lim_{x\to 0}(|\cos x + \sin 2x + \sin 3x| - 1)\cot x$$
$$= \lim_{x\to 0}(\cos x + \sin 2x + \sin 3x - 1)\cot x$$
$$= \lim_{x\to 0} \frac{\cos x - 1 + \sin 2x + \sin 3x}{x \cdot \frac{\tan x}{x}}$$
$$= \lim_{x\to 0}\left(\frac{\cos x - 1}{x} + \frac{\sin 2x}{x} + \frac{\sin 3x}{x}\right) = 2 + 3 = 5$$

Q64

Matrices value of whose determinant is zero.

$$\underbrace{\begin{bmatrix}2 & 3\\ 2 & 3\end{bmatrix}}_{2}\underbrace{\begin{bmatrix}2 & 2\\ 3 & 3\end{bmatrix}}_{2}\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}\begin{bmatrix}3 & 3\\ 3 & 3\end{bmatrix} \rightarrow 6$$

From the remaining 10 matrices, 5 have negative values of their determinant and 5 have positive values of their determinant.

$\therefore \quad$ Required probability $= \frac{5}{11}$

Q65

$\text{Let} \quad z = x + iy \quad \therefore \quad \bar{z} = x - iy$

$\therefore \quad (2iy)^2 = 12\left(x^2 + y^2\right) - 4 \Rightarrow 12x^2 + 16y^2 = 4$

$3x^2 + 4y^2 = 1 \quad \Rightarrow \quad \dfrac{x^2}{\frac{1}{3}} + \dfrac{y^2}{\frac{1}{4}} = 1$

$\therefore \quad x = \sqrt{\dfrac{1}{3}}\cos\theta, y = \sqrt{\dfrac{1}{4}}\sin\theta$

$\therefore \quad 3\sqrt{3}\,\mathrm{Re}(z) + 8\,\mathrm{Im}(z) = 3\cos\theta + 4\sin\theta$

$\therefore \quad \max = 5$

Q66

$\text{As, } \left(\alpha^2 + \sqrt{3}\right) = -(3)^{1/4} \cdot \alpha$

$\Rightarrow \left(\alpha^4 + 2\sqrt{3}\alpha^2 + 3\right) = \sqrt{3}\alpha^2 \text{ (On squaring)}$

$\therefore \left(\alpha^4 + 3\right) = (-)\sqrt{3}\alpha^2$

$\Rightarrow \alpha^8 + 6\alpha^4 + 9 = 3\alpha^4 \quad \text{(Again squaring)}$

$\therefore \alpha^8 + 3\alpha^4 + 9 = 0$

$\Rightarrow \alpha^8 = -9 - 3\alpha^4$

(Multiply by α^4)

So, $\alpha^{12} = -9\alpha^4 - 3\alpha^8$

$\therefore \quad \alpha^{12} = -9\alpha^4 - 3\left(-9 - 3\alpha^4\right)$

$\Rightarrow \alpha^{12} = -9\alpha^4 + 27 + 9\alpha^4$

Hence, $\alpha^{12} = (27)^2$

$\Rightarrow \left(\alpha^{12}\right)^8 = (27)^8$

$\Rightarrow \alpha^{96} = (3)^{24}$

Similarly $\beta^{96}=(3)^{24}$

$\therefore \alpha^{96}\left(\alpha^{12}-1\right)+\beta^{96}\left(\beta^{12}-1\right)=(3)^{24}\times 52$

$\Rightarrow$ Option (3) is correct.

Q67

$|[\vec{\mathbf{a}}\,\vec{\mathbf{b}}\,\vec{\mathbf{c}}]|=30$

$|abc\sin\theta\cos\phi|=30\Rightarrow\theta=\dfrac{\pi}{2},\phi=0\Rightarrow\vec{\mathbf{a}},\vec{\mathbf{b}},\vec{\mathbf{c}}$ are mutually perpendicular

$$\begin{aligned}&(2\vec{\mathbf{a}}+\vec{\mathbf{b}}+\vec{\mathbf{c}})\cdot[(\vec{\mathbf{a}}\times\vec{\mathbf{c}})\times(\vec{\mathbf{a}}-\vec{\mathbf{c}})+\vec{\mathbf{b}}]\\&=(2\vec{\mathbf{a}}+\vec{\mathbf{b}}+\vec{\mathbf{c}})\cdot\left[(\vec{\mathbf{a}}\cdot\vec{\mathbf{a}})\vec{\mathbf{c}}+c^2\vec{\mathbf{a}}+\vec{\mathbf{b}}\right)\\&=2a^2c^2+b^2+a^2c^2=3a^2c^2+b^2=300+9=309\end{aligned}$$

$$\therefore\quad\frac{k}{103}=\frac{309}{103}=3$$

Q68

$P(x,y),Q(1,0)$

$PQ=\sqrt{(x-1)^2+y^2}$

$PQ^2=x^2+1-2x+y^2$

$PQ^2=x^2-2x+1+2x^3-3x^2+9$

$PQ^2=2x^3-2x^2-2x+10$

$\frac{d\left(PQ^2\right)}{dx}=6x^2-4x-2=0$

$6x^2-4x-2=0$

$3x^2-2x-1=0$

$3x^2-3x+x-1=0$

$3x(x-1)+(x-1)=0$

$(x-1)(3x+1)=0$

$x=1,x=-\frac{1}{3}$

$\frac{d^2\left(PQ^2\right)}{dx^2}=12x-4$

So,PQ^2 is minimum at $x = 1y^2 = 2 + 9 - 3 = 8$

$\left(PQ\right)_{\text{minimum}} = \sqrt{0+8} = 2\sqrt{2}$

Q69

$\det.(\text{adj}(\text{adj}(\text{adj}(\text{adj}A)))) = |A|^{(3-1)^4}$

$= |A|^{16} = 4^8.5^{16}$

$\Rightarrow |A| = \pm 10$

$$|A| = \begin{vmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{vmatrix} = x + y + z = \pm 10$$

$\because x, y, z \in N \Rightarrow x + y + z = -10$ (not possible)

Hence, $x + y + z = 10$

The number of such matrices $= {}^9C_2$

$= 36$

Q70

Let the general equation of circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$ ..... (i)

It cuts the circle $x^2 + y^2 - 20x + 4 = 0$ orthogonally

$\therefore 2(-10g + 0 \times f) = c + 4 \Rightarrow -20g = c + 4$ .....(ii)

$\because$ circle (i) touches $x = 2$

therefore, perpendicular distance from centre to the tangent to the circle=radius

$\Rightarrow \left|\frac{-g+0-2}{\sqrt{1^2+0^2}}\right| = \sqrt{g^2 + f^2 - c}$

$\Rightarrow (g+2)^2 = g^2 + f^2 - c$

$\Rightarrow g^2 + 4 + 4g = g^2 + f^2 - c \Rightarrow 4g + 4 = f^2 - c$ ... (iii)

on eliminating c from (ii) and (iii) we get

$-16g + 4 = f^2 + 4 \Rightarrow f^2 + 16g = 0$

Hence, locus of $(-g, -f)$ is,

$y^2 - 16x = 0$ (replacing $-f$ & $-g$ by x & y)

Q71

Let $a = 2x + 1,\ b = 2y + 1, c = 2z + 2$, where x, y, z $\in$ whole number

$\therefore \quad a + b + c = 16$

$\Rightarrow \quad 2x + 1 + 2y + 1 + 2z + 2 = 16$

$\Rightarrow \quad x + y + z = 6$

$\Rightarrow$ Number of integral solutions

$= {}^{6+3-1}C_{3-1} = {}^{8}C_{2} = 28$

Q72

$T_{r+1} = {}^{n}C_r 2^r x^r$

$a_k = {}^{n}C_k 2^k$

$$\sum_{k=0}^{n}(3k+1)^{n}C_k \cdot a_k = 3\sum_{k=0}^{n} k \cdot {}^{n}C_k \cdot 2^k + \sum_{k=0}^{n} {}^{n}C_k \cdot 2^k = 3\sum_{k=0}^{n} {}^{n\cdot n-1}C_{k-1} \cdot 2^k + (1+2)^n$$

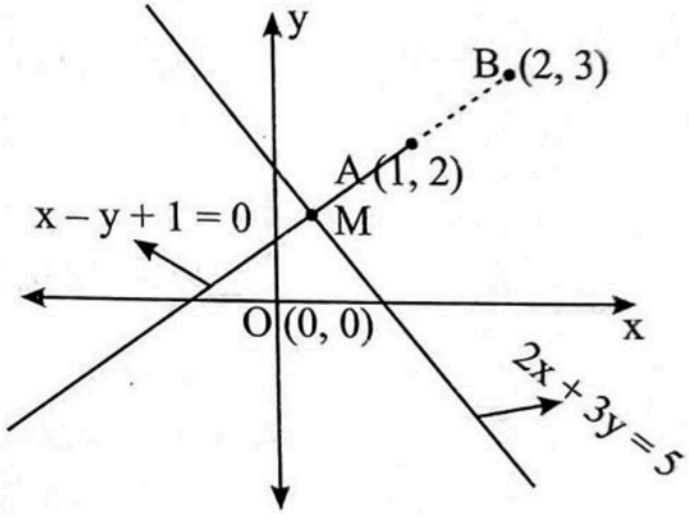
$$= 3n\sum_{k=0}^{n} {}^{n-1}C_{k-1} \cdot 2^{k+1} + 3^n = 2 \cdot 3n(1+2)^{n-1} + 3^n = 2n \cdot 3^n + 3^n = (2n+1)3^n$$

$\therefore p = 2, q = 1, r = 3$

Q73

As, $|MA - MB| \leq AB$

$\therefore \quad |MA - MB|_{\text{maximum}} = AB$, which is possible when M is the point of intersection of line joining



$A(1, 2), B(2, 3)$ and $2x + 3y = 5$

. So, $M\left(x = \frac{2}{5}, y = \frac{7}{5}\right)$

Q74

The system of equations has a non-trivial solution if and only if $\begin{vmatrix} \sin 3\theta & -2 & 3 \\ \cos 2\theta & 8 & -7 \\ 2 & 14 & -11 \end{vmatrix} = 0$

Applying $R_2 \rightarrow R_2 + 4R_1$, $R_3 \rightarrow R_3 + 7R_1$, we get,

$$\begin{vmatrix} \sin 3\theta & -2 & 3 \\ \cos 2\theta + 4\sin 3\theta & 0 & 5 \\ 2 + 7\sin 3\theta & 0 & 10 \end{vmatrix} = 0$$

Expanding along C_2, we get,

$2(\cos 2\theta + 4\sin 3\theta) - (2 + 7\sin 3\theta) = 0$

$\Rightarrow 2 - 2\cos 2\theta - \sin 3\theta = 0$

$\Rightarrow 4\sin^2\theta - (3\sin\theta - 4\sin^3\theta) = 0$

$\Rightarrow \sin\theta\,(4\sin^2\theta + 4\sin\theta - 3) = 0$

$\Rightarrow \sin\theta\,(2\sin\theta - 1)\,(2\sin\theta + 3) = 0$

$\Rightarrow \sin\theta = 0$ or $\sin\theta = 1/2$.

[$\sin\theta = -3/2$ is not possible]

$\therefore$ For, $\theta = n\pi$ the system of equations has a non-trivial solution.

Q75

Mean $= 6 = \frac{a+b+8+5+10}{5} \Rightarrow a + b = 7$

Variance $= 6.8 = \frac{a^2+b^2+64+25+100}{5} - 36$

$\Rightarrow a^2 + b^2 = 25$

$2ab = (a+b)^2 - (a^2+b^2) \Rightarrow ab = 12$

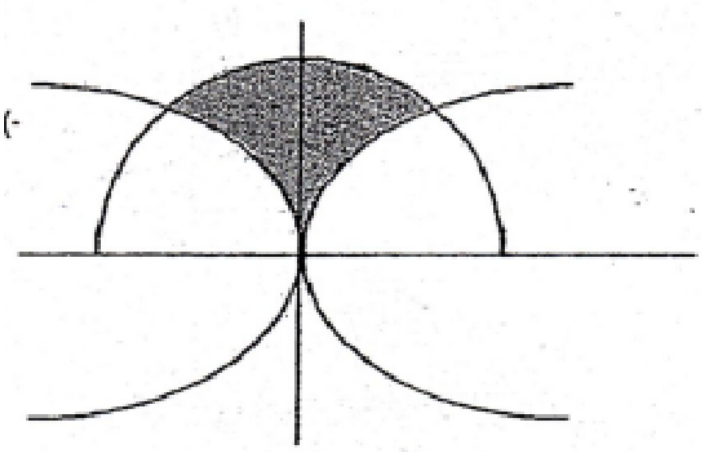
Now, $(a^3 + b^3) = (a+b)^3 - 3ab(a+b)$

$= 343 - 3 \times 12 \times 7$

$= 91$

Q76

Required area $= 2\int_0^1 \left(\sqrt{4-x^2} - \sqrt{3}x\right) dx$



$$= 2\left(\frac{x}{2}\sqrt{4-x^2} + \frac{4}{2}\sin^{-1}\left(\frac{x}{2}\right) - \frac{\sqrt{3}\cdot 2x^{1/2}}{3}\right)_0^1$$

$$= \frac{2\pi - \sqrt{3}}{3}$$

Q77

Given,

$$xdy = \left(\sqrt{x^2+y^2} + y\right)dx$$

$$\Rightarrow xdy - ydx = \sqrt{x^2+y^2}dx$$

$$\Rightarrow \frac{xdy - ydx}{x^2} = \sqrt{1+\frac{y^2}{x^2}} \cdot \frac{dx}{x}$$

$$\Rightarrow \frac{d\left(\frac{y}{x}\right)}{\sqrt{1+\left(\frac{y}{x}\right)^2}} = \frac{dx}{x}$$

Now integrating both side we get,

$$\Rightarrow \int \frac{d\left(\frac{y}{x}\right)}{\sqrt{1+\left(\frac{y}{z}\right)^2}} = \int \frac{dx}{x}$$

$$\Rightarrow \ln\left(\frac{y}{x} + \sqrt{\left(\frac{y}{x}\right)^2 + 1}\right) = \ln x + \ln c$$

$$\Rightarrow \frac{y + \sqrt{y^2+x^2}}{x} = cx$$

$$\Rightarrow y + \sqrt{y^2+x^2} = cx^2$$

Now given when $x=1, y=0 \Rightarrow 0+1=c \Rightarrow c=1$

So equation of curve is $y+\sqrt{x^2+y^2}=x^2$

Q78

(A) $p \vee(p \Rightarrow q)$ is false only if p is false and $(p \Rightarrow q)$ is false which is never possible simultaneously. Hence, $p \vee(p \Rightarrow q)$ is a tautology

(C) $p \Rightarrow(p \vee q)$ is false only if p is true and $(p \vee q)$ is false which is never possible simultaneously. Hence, $p \Rightarrow(p \vee q)$ is a tautology

(B) $p \Rightarrow(p \wedge q)$ is false only if p is true & $p \wedge q$ is false which is possible when p is true & q is false. Hence, $p \Rightarrow(p \wedge q)$ is not a tautology

Q79

The required number is of the form $2(2K+1)$, i.e. it will contain exactly one power of 2.

Also, we have to exclude 2 as $K \in N$.

Number of required divisors

$=1 \cdot 5 \cdot 11 \cdot 7-1=385-1=384$

Q80

Here, a,b,c are positive real numbers, so we can use AM GM inequality

$$\frac{\frac{a}{3}+\frac{a}{3}+\frac{a}{3}+\frac{b}{2}+\frac{b}{2}+c}{6} \geq\left(\frac{a^3}{27} \cdot \frac{b^2}{4} \cdot c\right)^{1/6}$$

$$\Rightarrow\left(\frac{a^3}{27} \cdot \frac{b^2}{4} \cdot c\right) \leq 1 \Rightarrow a^3 b^2 c \leq 3^3 \cdot 2^2$$

Q81

Making the substitution $x \mapsto \pi-x$ yields $I(a)=\int_0^\pi \log\left(1+2a\cos x+a^2\right)dx=I(-a)$ so that

$I(a)=I(-a). \quad (\dagger)$

Then, consider

$$\begin{aligned} I(a)+I(-a) &= \int_0^{\pi} \log\left(\left(1-2a\cos x+a^2\right)\left(1+2a\cos x+a^2\right)\right)dx \\ &= \int_0^{\pi} \log\left(\left(1+a^2\right)^2-(2a\cos x)^2\right)dx \end{aligned}$$

Using double angle formulae produces

$$\begin{aligned} I(a)+I(-a) &= \int_0^{\pi} \log\left(1+2a^2+a^4-2a^2(1+\cos 2x)\right)dx \\ &= \int_0^{\pi} \log\left(1-2a^2\cos 2x+a^4\right)dx, \end{aligned}$$

so we may let $x \mapsto \frac{1}{2}x$ to give

$I(a)+I(-a) = \frac{1}{2}\int_0^{2x} \log\left(1-2a^2\cos x+a^4\right)dx$

We can then split the integral at π and set $x \mapsto 2\pi - x$ for the second integral:

$$\begin{aligned} I(a)+I(-a) &= \frac{1}{2}I\left(a^2\right)+\frac{1}{2}\int_{\pi}^{2\pi} \log\left(1-2a^2\cos x+a^4\right)dx \\ &= \frac{1}{2}I\left(a^2\right)+\frac{1}{2}\int_0^{\pi} \log\left(1-2a^2\cos x+a^4\right)dx \\ &= I\left(a^2\right). \end{aligned}$$

We thus have (applying (†))

$I(a) = \frac{1}{2}I\left(a^2\right).$

Q82

Let

$f(x) = x^5 - x^3 + x - 2$

$f'(x) = 5x^4 - 3x^2 + 1 > 0 \forall x \in R$

$\therefore f(x)$ is increasing $\Rightarrow$ only one real root.

$f(1) = -1, f(2) = 24 \Rightarrow 1 < \alpha < 2$

Since, α is a root of $x^5 - x^3 + x - 2 = 0$

$$\begin{aligned} \Rightarrow \alpha^5 - \alpha^3 + \alpha &= 2 \\ \alpha^4 - \alpha^2 + 1 &= \frac{2}{\alpha} \end{aligned}$$

$$\left(\alpha^2+1\right)\left(\alpha^4-\alpha^2+1\right) = \frac{2}{\alpha}\left(\alpha^2+1\right)$$

$$\alpha^6+1=2\alpha+\frac{2}{\alpha}\Rightarrow\alpha^6=2\alpha+\frac{2}{\alpha}-1$$

$$g(\alpha)=2\alpha+\frac{2}{\alpha}-1$$

$$g'(\alpha)=2-\frac{2}{\alpha^2}=\frac{2}{\alpha^2}\left(\alpha^2-1\right)=\frac{2}{\alpha^2}(\alpha-1)(\alpha+1)$$

g is increasing for $\alpha>1$.

$$g(1)<g(\alpha)<g(2)$$

$$3<g(\alpha)<4$$

$$3<\alpha^6<4$$

$$\left[\alpha^6\right]=3$$

$$\therefore\quad\left[\alpha^6\right]=3$$

Q83

As $(3053)^{456}-(2417)^{333}=(339\times9+2)^{456}-(269\times9-4)^{333}$

Remainder of given number is same as remainder of $2^{456}+4^{333}$ and $2^{456}+4^{333}=(64)^{76}+(64)^{111}$

$=(1+63)^{76}+(1+63)^{111}=(1+9\times7)^{76}+(1+9\times7)^{111}$

Hence the remainder is 2 .

Q84

$$S_n=1!+2!+3!+4!+5!+6!+7!+\ldots\ldots$$

$$S_n=1+2+6+24+120+720+7I\qquad\text{...(1)}$$

$$S_n=873+7I$$

$$\frac{S_n}{7}=\frac{873}{7}+I$$

$$\left[\frac{S_n}{7}\right]=124+I\quad\text{... (2)}$$

$$7\left[\frac{S_n}{7}\right]=868+7I$$

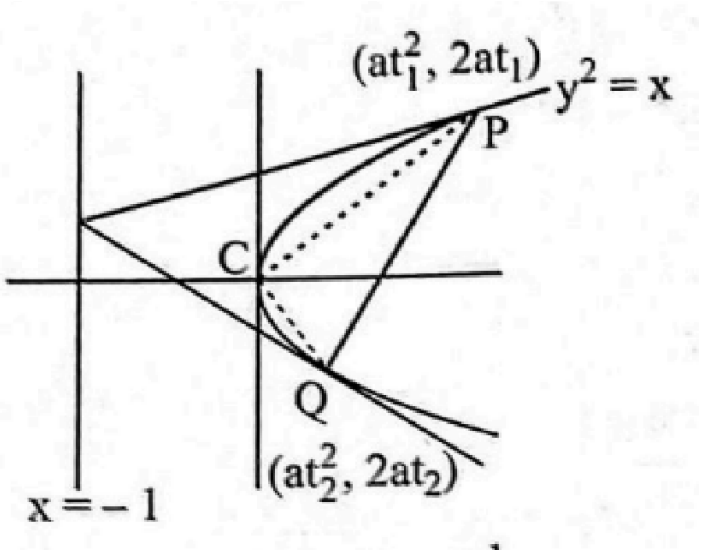
Eqn. (1) - Eqn. (2) $=5$

$$T=\sin^{-1}(\sin5)=5-2\pi$$

$$\int_0^1\frac{5-2\pi}{\sqrt{1-x^2}}dx=\frac{(5-2\pi)\pi}{2}=\frac{5\pi}{2}-\pi^2$$

$$\left(\frac{b}{c}+a\right)=\frac{2}{2}+5=6$$

Q85



$$at_1t_2 = -1 \Rightarrow t_1t_2 = \frac{-1}{a} = -4$$

$$\Delta = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ at_1^2 & 2at_1 & 1 \\ at_2^2 & 2at_2 & 1 \end{vmatrix} = \frac{1}{2}\left(2a^2t_1^2t_2 - 2a^2t_1t_2^2\right)$$

$$\Delta = \left|a^2t_1t_2\left(t_1 - t_2\right)\right| = \frac{1}{4}|t_1 - t_2|$$

$$\Delta = \frac{1}{4}\left(t_1 + \frac{4}{t_1}\right) = \frac{1}{4}\left[\left(\sqrt{t_1} - \frac{2}{\sqrt{t_1}}\right)^2 + 4\right]$$

$$\Delta_{\text{min.}} = 1 = M$$

$$4M = 4$$

Q86

$$S = \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{2}{9}\right) + \tan^{-1}\left(\frac{1}{8}\right) + \tan^{-1}\left(\frac{2}{25}\right) + \tan^{-1}\left(\frac{1}{18}\right) + \ldots\ldots\infty$$

$$= \tan^{-1}\left(\frac{2}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right) + \tan^{-1}\left(\frac{2}{16}\right) + \tan^{-1}\left(\frac{2}{25}\right) + \tan^{-1}\left(\frac{2}{36}\right) + \ldots\ldots\infty$$

$$T_n = \tan^{-1}\left(\frac{2}{(n+1)^2}\right)$$

$$S = \sum_{r=1}^{\infty}\tan^{-1}\left(\frac{2}{r^2+2r+1}\right) = \sum_{r=1}^{\infty}\tan^{-1}\left(\frac{(r+2)-r}{1+r(r+2)}\right) = \sum_{r=1}^{\infty}\tan^{-1}(r+2) - \tan^{-1}(r)$$

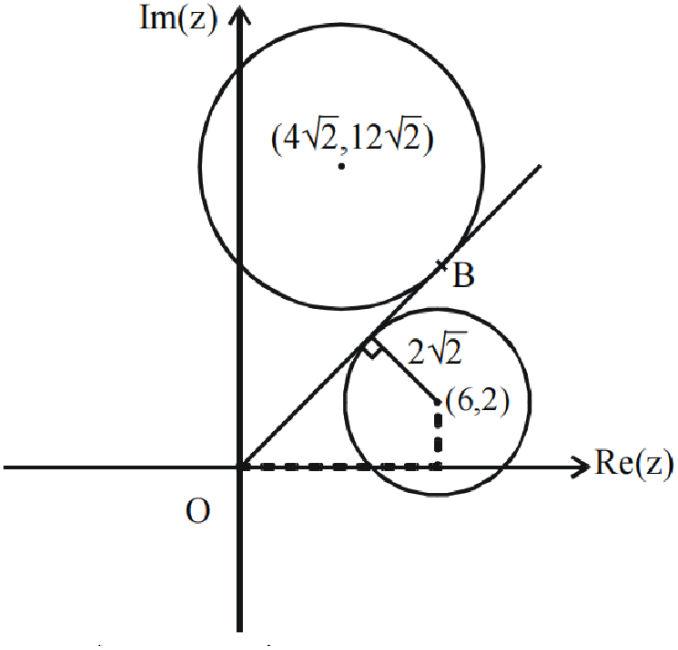
$$S = \lim_{n\to\infty}\tan^{-1}\left(\frac{3n^2+7n}{n^2+9n+10}\right) = \tan^{-1}(3)$$

(3) Let

Hence, $\tan S = \tan\left(\tan^{-1}3\right) = 3$

Q87

Let the equation of OAB be $y = mx$



$$\Rightarrow \left|\frac{6m-2}{\sqrt{1+m^2}}\right| = 2\sqrt{2} \Rightarrow \frac{(3m-1)^2}{(1+m^2)} = 2$$

$$\Rightarrow \frac{9m^2-6m+1}{1+m^2} = 2 \Rightarrow 7m^2-6m-1=0$$

$$\Rightarrow 7m^2-7m+m-1=0$$

$$7m(m-1)+(m-1)=0 \Rightarrow m=1$$

$\Rightarrow y = x$ must be tangent to the second circle

$$\Rightarrow \left|\frac{12\sqrt{2}-4\sqrt{2}}{\sqrt{2}}\right| = k \Rightarrow k = 8$$

Q88

$$f(x) = x + \sin x - [x+\sin x] + [x-\sin x] + [x]$$

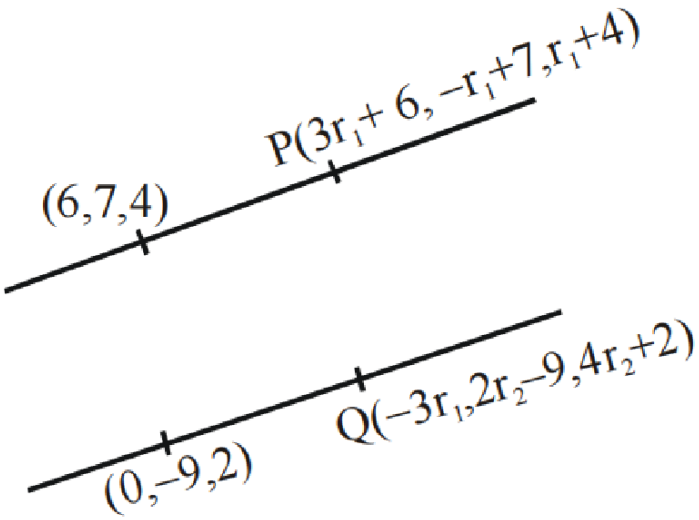
$x + \sin x = 0, 1, 2, 3 \Rightarrow x = 0, \alpha_1, \alpha_2, \alpha_3$

$x - \sin x = 0, 1, 2, 3 \Rightarrow x = 0, \beta_1, \beta_2, \beta_3$

f is continuous at $x = 0$, but discontinuous at $\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3, 1, 2, 3$

$\therefore$ Number of points of discontinuity $= 9$.

Q89



direction ratio of

$PQ \equiv (-3r_2 - 3r_1 - 6, 2r_2 + r_1 - 16, 4r_2 - r_1 - 2)$

Now, $-3(-3r_2 - 3r_1 - 6) + 2(2r_2, r_1 - 16)$

$+ 4(4r_2 - r_1 - 2) = 0$

$7r_1 + 29r_2 = 22 \quad \ldots(1)$

$3(-3r_2 - 3r_1 - 6) - (2r_2 + r_1 - 16) + (4r_2 - r_1 - 2) = 0$

$7r_1 + 29r_2 = 22 \quad \ldots(4r_2 - r_1 - 2) = 0$

$3(-3r_2 - 3r_1 - 6) - (2r_2 + r_1 - 16) + (2)$

$11r_1 + 7r_2 = -4$

On solving (1) & (2)

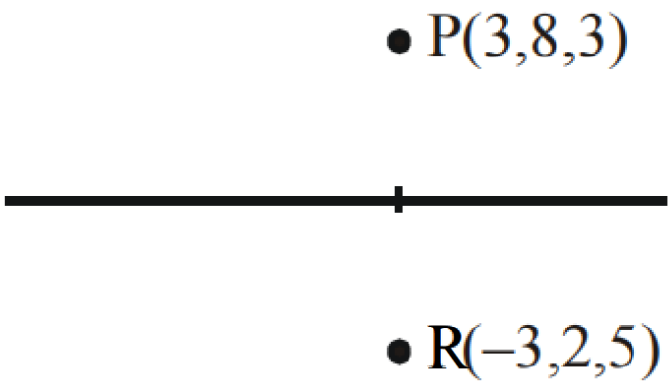
$r_1 = -1, r_2 = 1$

so $P(3, 8, 3)$

Image of $P(3, 8, 3)$ w.r.t Plane $3x + 3y - z = 11$

$R(-3, 2, 5)$

$a + b + c = 4$



Q90

$T_1 = (\vec{\mathbf{a}} \times \vec{\mathbf{b}}) \times (\vec{\mathbf{c}} \times \vec{\mathbf{d}})$

$T_2 = (\vec{\mathbf{a}} \times \vec{\mathbf{c}}) \times (\vec{\mathbf{d}} \times \vec{\mathbf{b}})$

$T_3 = (\vec{\mathbf{a}} \times \vec{\mathbf{d}}) \times (\vec{\mathbf{b}} \times \vec{\mathbf{c}})$

$$T_1 = \left[\vec{\mathbf{c}} \quad \vec{\mathbf{d}} \quad \vec{\mathbf{a}}\right]\vec{\mathbf{b}} - \left[\vec{\mathbf{c}} \quad \vec{\mathbf{d}} \quad \vec{\mathbf{b}}\right]\vec{\mathbf{a}}$$

$$\text{Then, } T_2 = \left[\vec{\mathbf{d}} \quad \vec{\mathbf{b}} \quad \vec{\mathbf{a}}\right]\vec{\mathbf{c}} - \left[\vec{\mathbf{d}} \quad \vec{\mathbf{b}} \quad \vec{\mathbf{c}}\right]\vec{\mathbf{a}}$$

$$T_3 = \left[\vec{\mathbf{a}} \quad \vec{\mathbf{d}} \quad \vec{\mathbf{c}}\right]\vec{\mathbf{b}} - \left[\vec{\mathbf{a}} \quad \vec{\mathbf{d}} \quad \vec{\mathbf{b}}\right]\vec{\mathbf{c}}$$

$$\text{Sum } = -2\left[\vec{\mathbf{b}} \quad \vec{\mathbf{c}} \quad \vec{\mathbf{d}}\right]\vec{\mathbf{a}} + k\vec{\mathbf{a}} = 0$$

$$\left[\text{ Given}\left[\vec{b} \quad \vec{\mathbf{c}} \quad \vec{d}\right] = 48\right]$$

$$\text{Hence, } \quad k = 96$$